

General Relativity

Sohrab Maleki

Autumn 1404

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Introduction

This content was written with inspiration from the course “Introduction to General Relativity” offered by Dr. Khosravi in autumn 1404. Note that this writing is not the course notes. I have simply written the theory in a way that I think follows a logical progression (which also partly agrees with the course content). At the beginning of each chapter, I have written questions that I have no answer to (or questions that can be asked, whose answers I have not addressed in this text).

The prerequisites that are needed and that I use but do not introduce are these:

- The Lagrangian and Hamiltonian formulations of classical mechanics
- Classical electromagnetism (Maxwell)
- Special relativity and covariant electromagnetism
- Definitions and transformations of tensors

Throughout this text, I take the Minkowski metric to be $(-+++)$, and the Einstein summation convention is used (unless stated otherwise).

Chapter 1

Scalar Field

Questions:

1. What is the necessary and sufficient condition on the dynamics of a system for a Lagrangian to exist? How can one tell whether the dynamics of a system can even be expressed in the Lagrangian formalism?
2. I could not precisely make the discrete Lagrangian continuous in such a way that a dependence on $\partial\phi/\partial x$ appears in it. But I think this can be done with a little care. If you managed to do it, I would be happy if you told me too!
3. Why do we not use the Hamiltonian formalism? (Probably just for mathematical convenience; but if you have a better answer, you can share it with me.)
4. Why do we only take the Lagrangian to be a function of first-order derivatives? The Galileon has a second-order derivative but its equation of motion turns out to be second order.
5. I would be happier to see a more precise formulation of the ghosts we were getting rid of.

1.1 From Classical Mechanics to a Classical Field

It is clear that Newton's equations can *necessarily* be written in the Lagrangian formalism. Now consider a one-dimensional nonrelativistic system made up of identical masses and springs connected to one another. I write the Lagrangian of this system:

$$L(\{\phi_n\}, \{\dot{\phi}_n\}) = \frac{1}{2}m \sum_n \dot{\phi}_n^2 - \frac{1}{2}k \sum_n (\phi_n - \phi_{n-1})^2 \quad (1.1)$$

where ϕ_n is the displacement of the n th body from equilibrium. I write the equations of motion (Euler-Lagrange):

$$\frac{\partial L}{\partial \phi_n} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}_n} \right) = 0 \quad (1.2)$$

A simple substitution gives:

$$m\ddot{\phi}_n = -k \left(2\phi_n(t) - \phi_{n-1}(t) - \phi_{n+1}(t) \right)$$

The term in parentheses is proportional to the discretized form of the second derivative. To show this, I define:

$$\Delta[f(\phi_n)] := \frac{f(\phi_n) - f(\phi_{n-1})}{\delta x}$$

where δx is the distance between two consecutive masses at equilibrium. This is the discretized form of the derivative. Applying this twice gives:

$$\begin{aligned} \Delta[\Delta[f(\phi_n)]] &= \Delta\left[\frac{f(\phi_n) - f(\phi_{n-1})}{\delta x}\right] \\ &= \frac{f(\phi_n) - f(\phi_{n-1}) - f(\phi_{n-1}) + f(\phi_{n-2})}{\delta x^2} \end{aligned}$$

So:

$$\Delta[\Delta[f(\phi_{n+1})]] = \frac{f(\phi_{n+1}) - 2f(\phi_n) + f(\phi_{n-1}))}{\delta x^2}$$

Therefore, the equation of motion takes the following form:

$$m\ddot{\phi}_n(t) = k\delta x^2 \Delta[\Delta[\phi_{n+1}(t)]]$$

Now I want to examine the continuum limit of this equation. The continuum limit means $\delta x \rightarrow 0$, but in this case, some of the n 's go to infinity. So I define:

$$x := n\delta x$$

and from now on, instead of labeling the ϕ 's with n , I label them with x . I rewrite the equation:

$$\begin{aligned} \frac{m}{\delta x} \ddot{\phi}(x, t) &= k\delta x \Delta[\Delta[\phi(x + \delta x, t)]] \\ \mu \ddot{\phi}(x, t) &= \kappa \Delta[\Delta[\phi(x + \delta x, t)]] \end{aligned}$$

where:

$$\mu := \frac{m}{\delta x}, \quad \kappa := k\delta x$$

Now, keeping μ, κ fixed (rather than m, k), I let δx go to zero. Note that from the definition of the operation Δ , one can see that:

$$\lim_{\delta x \rightarrow 0} \Delta[\phi(x + \delta x, t)] = \frac{\partial \phi(x, t)}{\partial x}$$

So:

$$\lim_{\delta x \rightarrow 0} \Delta[\Delta[\phi(x + \delta x, t)]] = \frac{\partial^2 \phi(x, t)}{\partial x^2}$$

where continuity has also been used in these results. Finally, we will have:

$$v^2 \frac{\partial^2 \phi}{\partial x^2} - \frac{\partial^2 \phi}{\partial t^2} = 0 \quad v := \sqrt{\kappa/\mu}$$

which is a wave equation with a constant speed v .

Now I want to write the continuum limit of the Lagrangian. First, I rewrite the Lagrangian as follows:

$$L[\phi(x), \dot{\phi}(x)] = \frac{1}{2} \mu \sum_x \dot{\phi}(x)^2 \delta x - \frac{1}{2} \kappa \sum_x \frac{1}{\delta x} (\Delta[\phi(x)] \delta x)^2$$

Clearly, its continuum limit (that is, $\delta x \rightarrow 0$) takes the following form:

$$L[\phi(x), \dot{\phi}(x)] = \int \left[\frac{1}{2} \mu \dot{\phi}^2 - \frac{1}{2} \kappa \left(\frac{\partial \phi}{\partial x} \right)^2 \right] dx$$

I define the Lagrangian density:

$$\mathcal{L}[\phi(x), \dot{\phi}(x)] := \frac{1}{2} \mu \dot{\phi}^2 - \frac{1}{2} \kappa \left(\frac{\partial \phi}{\partial x} \right)^2$$

So the action becomes:

$$S = \iint \mathcal{L}[\phi(x), \dot{\phi}(x)] dx dt$$

It seems that everything is fine. All the expressions turned into a number of convergent integrals (in fact, I defined μ, κ in such a way that these would converge), and I wrote the continuous theory. Clearly, making the Lagrangian continuous was much more straightforward than making the equations of motion continuous. A question comes up here: *Can one make the Lagrangian continuous and derive the equations of motion from it?*

If you look carefully, you see that directly using the continuum limit of (1.2) is not possible. That is, to use that equation, one must discretize again, write the equation, and make it continuous again. This problem can also be understood from the fact that the continuum limit of the Lagrangian has no separate dependence on $\partial\phi/\partial x$. Here I make an assumption that is not obvious (but later I show that, for this example, it makes no difference). And that is that *one can turn this same Lagrangian $L[\phi(x), \dot{\phi}(x)]$ into $L[\phi, \dot{\phi}, \partial\phi/\partial x]$ without changing its expression, and calculate the equations of motion from the continuous Euler-Lagrange equations.*

1.2 Euler-Lagrange Equations of the Scalar Field

Here I take ϕ to be a general scalar field and derive Euler-Lagrange for it.

1.2.1 Lagrangian with First-Order Derivatives

I consider a Lagrangian density of the following form:

$$\mathcal{L}(\phi, \partial_\mu \phi, x^\mu)$$

I want to extremize the action over paths from x_A to x_B that agree at these two points:

$$S = \int d^4x \mathcal{L}(\phi, \partial_\mu \phi, x^\mu)$$

Now, if we change the extremal path a little, the action should not change to first order:

$$\delta S = \int d^4x \delta \mathcal{L}(\phi, \partial_\mu \phi, x^\mu) = 0$$

So:

$$0 = \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\mu \delta \phi \right)$$

On the other hand:

$$\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\mu \delta \phi = \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \phi \right) - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \right) \delta \phi$$

So:

$$0 = \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \right) \right) \delta \phi$$

for every $\delta \phi$. So:

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \right) = 0}$$

Note that the total derivative term vanishes because of the four-dimensional Stokes theorem:

$$\int_{\Omega} d^n x \partial_\mu f(x) = \int_{\partial \Omega} dS n_\mu f(x)$$

and for this to vanish, $\delta \phi$ must also vanish at infinity. It is clear that any total derivative term in the Lagrangian disappears in the same way and has no effect on the equation of motion. I define an equivalence class as follows:

$$\mathcal{L}(\phi, \partial_\mu \phi, x^\mu) \sim \mathcal{L}'(\phi, \partial_\mu \phi, x^\mu) \iff \exists F(\phi, \partial_\mu \phi, x^\mu) : \mathcal{L} - \mathcal{L}' = \frac{d}{dx^\mu} F^\mu(\phi, \partial_\mu \phi, x^\mu)$$

Since all Lagrangians in one equivalence class give the same equation of motion, I take these to be equivalent for the rest of this note. In such cases, instead of equality ($=$), I use the equivalence symbol ($\equiv$).

1.2.2 Lagrangian with Second-Order Derivatives

Now I consider the following Lagrangian:

$$\mathcal{L}(\phi, \partial_\mu \phi, \partial_\mu \partial_\nu \phi, x)$$

The method is the same as in the previous part. For simplicity, I only write the terms added to the previous method:

$$\begin{aligned} 0 &= \int d^4x \left(\cdots + \frac{\partial \mathcal{L}}{\partial \partial_\mu \partial_\nu \phi} \partial_\mu \partial_\nu \delta \phi \right) \\ &= \int d^4x \left(\cdots - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \partial_\nu \phi} \right) \partial_\nu \delta \phi \right) \\ &= \int d^4x \left(\cdots + \frac{d}{dx^\mu} \frac{d}{dx^\nu} \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \partial_\nu \phi} \right) \right) \delta \phi \end{aligned}$$

So the equation of motion turns into:

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \frac{d}{dx^\mu} \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \right) + \frac{d}{dx^\mu} \frac{d}{dx^\nu} \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \partial_\nu \phi} \right) = 0}$$

Higher-order derivatives can also be written in the same way.

1.3 Hamilton's Equations of the Scalar Field

In this section, I generalize the Hamiltonian method to fields. There are two methods for this:

1. Separating time evolution from spatial evolution and using the usual Hamiltonian method in mechanics. I call this method the “noncovariant form” of Hamilton's equations. Note that this does not mean it is necessarily noncovariant. Its form simply does not guarantee covariance.
2. Writing the method without separating time, at the cost of the geometry becoming multisymplectic¹ and the canonical momentum becoming a 4-vector. The Poisson bracket also has no obvious generalization in this method, but the form of the equations remains covariant.

1.3.1 Noncovariant Form

I define the conjugate momentum field of ϕ :

$$\pi(x) := \frac{\partial \mathcal{L}}{\partial \dot{\phi}}$$

The equations of motion (Euler–Lagrange) give:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \dot{\pi} - \nabla \cdot \frac{\partial \mathcal{L}}{\partial \nabla \phi} = 0$$

So:

$$\dot{\pi} = \frac{\partial \mathcal{L}}{\partial \phi} - \nabla \cdot \frac{\partial \mathcal{L}}{\partial \nabla \phi}$$

Now suppose that, during a small change in the field, its Lagrangian density changes by a small amount $d\mathcal{L}$. I write:

$$\begin{aligned} d\mathcal{L} &= \frac{\partial \mathcal{L}}{\partial \phi} d\phi + \frac{\partial \mathcal{L}}{\partial \dot{\phi}} d\dot{\phi} + \frac{\partial \mathcal{L}}{\partial \nabla \phi} \cdot d(\nabla \phi) + \nabla \mathcal{L} \cdot d\mathbf{x} + \frac{\partial \mathcal{L}}{\partial t} dt \\ &= \frac{\partial \mathcal{L}}{\partial \phi} d\phi + \pi d\dot{\phi} - \nabla \cdot \frac{\partial \mathcal{L}}{\partial \nabla \phi} d\phi + \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial \nabla \phi} d\phi \right) + \nabla \mathcal{L} \cdot d\mathbf{x} + \frac{\partial \mathcal{L}}{\partial t} dt \\ &= \dot{\pi} d\phi - \dot{\phi} d\pi + d(\pi \dot{\phi}) + \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial \nabla \phi} d\phi \right) + \nabla \mathcal{L} \cdot d\mathbf{x} + \frac{\partial \mathcal{L}}{\partial t} dt \end{aligned}$$

I define the Hamiltonian density:

$$\mathcal{H}(\pi, \phi; \mathbf{x}, t) := \pi \dot{\phi} - \mathcal{L}$$

¹multi-symplectic

Therefore:

$$d\mathcal{H} = \dot{\phi}d\pi - \dot{\pi}d\phi - \nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial \nabla \phi} d\phi \right) - \nabla \mathcal{L} \cdot d\mathbf{x} - \frac{\partial \mathcal{L}}{\partial t} dt$$

I define the Hamiltonian:

$$H(\pi, \phi; t) := \int d^3\mathbf{x} \mathcal{H}(\pi, \phi; \mathbf{x}, t)$$

It is clear that the Hamiltonian has no explicit spatial dependence. So:

$$dH = \int d^3\mathbf{x} \left(\dot{\phi}d\pi - \dot{\pi}d\phi - \frac{\partial \mathcal{L}}{\partial t} dt \right)$$

From the definition of the Hamiltonian, it is clear that:

$$dH[\phi, \pi] = \int d^3\mathbf{x} \left(\frac{\partial H}{\partial \phi} d\phi + \frac{\partial H}{\partial \pi} d\pi \right)$$

Comparison with the previous equation gives Hamilton's equations:

$$\frac{\partial H}{\partial \phi} = -\dot{\pi} \quad , \quad \frac{\partial H}{\partial \pi} = \dot{\phi}$$

As is clear, the equations in this method do not have a covariant form. But in principle, any covariant theory can also be written in this form.

1.3.2 Covariant Form

I define the conjugate momentum 4-vector of ϕ :

$$\pi^\mu(x) := \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi}$$

I write Euler-Lagrange:

$$\frac{\partial \mathcal{L}}{\partial \phi} = d_\mu \pi^\mu = \partial_\mu \pi^\mu$$

Now I write $d\mathcal{L}$:

$$\begin{aligned} d\mathcal{L} &= \frac{\partial \mathcal{L}}{\partial \phi} d\phi + \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} d(\partial_\mu \phi) + \partial_\mu \mathcal{L} dx^\mu \\ &= \partial_\mu \pi^\mu d\phi + \pi^\mu d(\partial_\mu \phi) + \partial_\mu \mathcal{L} dx^\mu \\ &= \partial_\mu \pi^\mu d\phi + d(\pi^\mu \partial_\mu \phi) - \partial_\mu \phi d\pi^\mu \end{aligned}$$

I define the Hamiltonian density:

$$\mathcal{H} := \pi^\mu \partial_\mu \phi - \mathcal{L}$$

So:

$$d\mathcal{H} = -\partial_\mu \pi^\mu d\phi + \partial_\mu \phi d\pi^\mu$$

So the equations of motion are obtained:

$$\frac{\partial \mathcal{H}}{\partial \phi} = -\partial_\mu \pi^\mu \quad , \quad \frac{\partial \mathcal{H}}{\partial \pi^\mu} = \partial_\mu \phi$$

These equations are automatically covariant, but the conjugate momentum of the scalar ϕ inevitably became a 4-vector. This makes the geometry of the problem more complicated and makes the generalization of the Poisson bracket to it less obvious. But none of these are clear reasons for the Hamiltonian formalism to be “defective.” Probably this method has no serious problem, but guessing the Lagrangian is simply easier than guessing the Hamiltonian.

Although quantum mechanics has forced us to learn how to construct Hamiltonians too; for example, the equation of a free field is:

$$\mathcal{H} = \frac{1}{2} \pi_\mu \pi^\mu$$

The covariant equation of motion of this Hamiltonian gives:

$$\partial_\mu \pi^\mu = 0$$

$$\partial_\mu \phi = \pi_\mu$$

where the first is conservation of energy and momentum, and the second is the definition of momentum in terms of the field. To understand better, I expand the spatial and time parts. I denote the zeroth component of the momentum 4-vector by p and its other three components by $\mathbf{j}$:

$$\frac{\partial p}{\partial t} + \nabla \cdot \mathbf{j} = 0$$

$$\frac{\partial \phi}{\partial t} = p$$

$$-\nabla \phi = \mathbf{j}$$

So:

$$\frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi = 0$$

which is the wave equation. If I had written the Hamiltonian in the form

$$\mathcal{H} = \frac{1}{2} \pi_\mu \pi^\mu + \mathcal{V}(\phi)$$

it would have given:

$$\frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi = -\mathcal{V}'(\phi)$$

Note that one cannot interpret the zeroth component of the 4-vector π^μ as energy and the remaining components as momentum. Also, the Hamiltonian density defined here (the covariant form) is not equal to the Hamiltonian density of the previous section (the noncovariant form). In fact, π^0 in this method is the same as π in the previous method.

1.3.3 Poisson Bracket

First I define the Poisson bracket for the noncovariant form, and then I show that the covariant form of the Hamiltonian does not generalize easily.

Consider an observable:

$$Q(\phi, \pi)$$

One can write:

$$\begin{aligned}\frac{d}{dt}Q &= \frac{\partial Q}{\partial \phi} \dot{\phi} + \frac{\partial Q}{\partial \pi} \dot{\pi} \\ &= \frac{\partial Q}{\partial \phi} \frac{\partial H}{\partial \pi} - \frac{\partial Q}{\partial \pi} \frac{\partial H}{\partial \phi} \\ &=: \{Q, H\}\end{aligned}$$

Now consider the covariant form of the Hamiltonian. Again, consider the observable:

$$Q(\phi, \pi^\mu)$$

One can write:

$$\frac{d}{dx^\mu}Q = \frac{\partial Q}{\partial \phi} \partial_\mu \phi + \frac{\partial Q}{\partial \pi^\nu} \partial_\mu \pi^\nu$$

On the other hand:

$$\frac{\partial \mathcal{H}}{\partial \phi} = -\partial_\mu \pi^\mu \quad , \quad \frac{\partial \mathcal{H}}{\partial \pi^\mu} = \partial_\mu \phi$$

So:

$$\frac{d}{dx^\mu}Q = \frac{\partial Q}{\partial \phi} \frac{\partial \mathcal{H}}{\partial \pi^\mu} + \frac{\partial Q}{\partial \pi^\nu} \partial_\mu \pi^\nu$$

As is clear, the second term cannot easily be written in terms of the Hamiltonian density so that the Poisson bracket can be defined from it.

1.4 Symmetry

In this section, I examine the symmetry constraint on field dynamics; I show Noether's theorem and its converse and write it for the two particular examples "symmetry under translation in spacetime" and "symmetry under rotation in spacetime" (corresponding to homogeneity and isotropy).

1.4.1 Noether's Theorem

Consider an arbitrary continuous² transformation such as $T_\epsilon : \phi \rightarrow \phi$. I define:

$$\delta_\epsilon := T_\epsilon - \mathbb{1}$$

Therefore:

$$T_\epsilon(\phi) = \phi + \delta_\epsilon \phi$$

I call the operation δ a symmetry of the system if and only if

$$\mathcal{L}(T_\epsilon(\phi), \partial_\mu T_\epsilon(\phi), x^\mu) \sim \mathcal{L}(\phi, \partial_\mu \phi, x^\mu)$$

Claim:

For every continuous symmetry, one can find a *conserved current* such as $j^\mu(x)$ such that:

$$\partial_\mu j^\mu(x) = 0$$

²A continuous transformation means that it can be brought arbitrarily close to the identity. For example, rotation and translation are continuous, but reflection is not continuous.

Proof: Consider the transformation T . For simplicity, I define:

$$\delta\mathcal{L} := \mathcal{L}(T_\epsilon(\phi), \partial_\mu T_\epsilon(\phi), x^\mu) - \mathcal{L}(\phi, \partial_\mu \phi, x^\mu)$$

From the definition of symmetry, I write:

$$\exists F(\phi, \partial_\mu \phi, x^\mu) : \delta_\epsilon \mathcal{L} = d_\mu F^\mu$$

From continuity, I know that for small ϵ , the left-hand side of this can be rewritten:

$$\frac{\partial \mathcal{L}}{\partial \phi} \delta_\epsilon \phi - d_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \right) \delta_\epsilon \phi + d_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta_\epsilon \phi \right) = d_\mu F^\mu$$

The first two terms are eliminated by the equations of motion. So:

$$d_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta_\epsilon \phi - F^\mu \right) = 0$$

So if I define:

$$\tilde{j}^\mu(\phi, \partial_\mu \phi, x^\mu) := \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta_\epsilon \phi - F^\mu$$

then:

$$d_\mu \tilde{j}^\mu = 0$$

Also, if I define:

$$j^\mu(x) := \tilde{j}^\mu(\phi(x), \partial_\mu \phi(x), x^\mu)$$

then:

$$\partial_\mu j^\mu = 0$$

□

This is a “conserved current.” If I represent the spatial and time parts of j^μ by ρ and $\mathbf{j}$ respectively, it follows that:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0$$

which is the local conservation equation of a quantity with density ρ and current $\mathbf{j}$.

Note that I used the continuity assumption where I assumed that by adjusting ϵ , one can make $\delta_\epsilon \phi$ arbitrarily small, and therefore I have only written its first-order terms.

1.4.2 Converse of Noether’s Theorem

I’ll write this later if I get the time

1.5 Energy-Momentum Tensor

Suppose the Lagrangian is invariant under translation in spacetime:

$$T_\epsilon[\phi(x)] = \phi(x^\mu + \epsilon^\mu)$$

For $\epsilon \ll 1$, one sees:

$$\begin{aligned}\delta_\epsilon \phi &= \epsilon^\mu \partial_\mu \phi \\ \delta \mathcal{L} &= \epsilon^\mu d_\mu \mathcal{L} = d_\mu (\epsilon^\mu \mathcal{L})\end{aligned}$$

The Noether current is obtained:

$$j^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \epsilon^\nu \partial_\nu \phi - \epsilon^\mu \mathcal{L}$$

I define:

$$T^\mu{}_\nu := \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi - \delta^\mu_\nu \mathcal{L}$$

I call this object the “energy-momentum tensor.” In this case, the conservation equation of the Noether current is obtained:

$$\partial_\mu T^\mu{}_\nu = 0$$

Also, in terms of the covariant form of the Hamiltonian:

$$T^\mu{}_\nu = \pi^\mu \frac{\partial \mathcal{H}}{\partial \pi^\nu} - \delta^\mu_\nu \left(\pi^\eta \frac{\partial \mathcal{H}}{\partial \pi^\eta} - \mathcal{H} \right)$$

For example, for the Klein–Gordon Lagrangian

$$\mathcal{L} = \frac{1}{2} \eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} m^2 \phi^2$$

there is symmetry under translation in spacetime. One can write its energy-momentum tensor:

$$T^\mu{}_\nu = \partial^\mu \phi \partial_\nu \phi - \delta^\mu_\nu \mathcal{L}$$

or, in other words:

$$T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - \eta^{\mu\nu} \mathcal{L}$$

Energy-Momentum Tensor of a Point Mass

Because I will use it later, I calculate the energy-momentum tensor of a point mass (rather than a field) here. From special relativity, I know that under symmetry under translation in spacetime, the momentum 4-vector is conserved:

$$P^\mu = (E/c, \mathbf{p})^T = mU^\mu$$

Consider a fluid in space. I write the spatial density of the momentum 4-vector:

$$T^{\mu 0}(x) := \rho U^\mu(x)$$

Its current is also obtained:

$$T^{\mu i} := \rho U^\mu(x) \frac{dx^i}{dt}$$

So I define the energy-momentum tensor in this way:

$$T^{\mu\nu} := \rho \gamma \dot{x}^\mu \dot{x}^\nu$$

From this definition, it is clear that the energy-momentum tensor is symmetric. Now I write it in this form:

$$T^{\mu\nu} = \rho(x) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{dt}$$

For a point particle, it gives:

$$\begin{aligned} T^{\mu\nu} &= m \delta^3(\mathbf{x} - \mathbf{x}(t)) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{dt} \\ &= m \int dt' \delta^4(\mathbf{x} - \mathbf{x}(t')) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{dt'} \\ &= m \int d\tau \delta^4(\mathbf{x} - \mathbf{x}(\tau)) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \end{aligned}$$

1.6 Examining a Few Terms in the Lagrangian

Here, I show a few particular terms and their effects in the Lagrangian. Since the equations of motion are linear, the effects of different terms of the Lagrangian are also added linearly to the equations of motion. Therefore, examining individual terms is useful.

First, I derive the effect of the term

$$\mathcal{L} = \partial_\mu \phi \partial^\mu \phi$$

in the equation of motion. This expression is also present in the “kinetic” term of the Klein–Gordon Lagrangian. Euler–Lagrange gives:

$$\partial_\mu \partial^\mu \phi = 0$$

So this term produces a second-order derivative in the equation of motion.

Now I examine different powers of this same term:

$$\mathcal{L} = (\partial_\mu \phi \partial^\mu \phi)^n = (\eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi)^n$$

I write the equations of motion:

$$\frac{d}{dx^\mu} (2n(\eta^{\gamma\delta} \partial_\gamma \phi \partial_\delta \phi)^{n-1} \eta^{\mu\nu} \partial_\nu \phi) = 2n \frac{d}{dx^\mu} ((\partial_\gamma \partial^\gamma \phi)^{n-1} \partial^\mu \phi) = 0$$

So:

$$2(n-1)(\partial_\gamma \partial^\gamma \phi)^{n-2} \partial_\mu \partial_\gamma \partial^\gamma \phi \partial^\mu \phi + (\partial_\gamma \partial^\gamma \phi)^{n-1} \partial_\gamma \partial^\gamma \phi \partial_\mu \partial^\mu \phi = 0$$

So:

$$2(n-1)(\partial_\gamma \partial^\gamma \phi) \partial_\mu \phi + (\partial_\gamma \partial^\gamma \phi)^2 \partial_\mu \phi = 0$$

It is clear that for every n , the equation of motion still remains second order.

As we have often seen in nature, *I assume that the equations of motion corresponding to one degree of freedom are second order.* In fact, I can also take this as the definition of a “degree of freedom”; because one can show that if the order of an equation of motion is even and greater than two, one can always define other fields whose equations of motion are each second order. Therefore,

the number of degrees of freedom really becomes equal to the number of fields. First, as an example, consider the following Lagrangian:

$$\mathcal{L} = (\partial_\mu \partial^\mu \phi)^2 = (\partial_\mu \partial^\mu \phi)(\partial_\nu \partial^\nu \phi) \equiv (\partial_\mu \partial^\nu \phi)(\partial_\nu \partial^\mu \phi)$$

I write its equation of motion:

$$d_\mu d_\nu \frac{\partial \mathcal{L}}{\partial \partial_\mu \partial_\nu \phi} = d_\mu d_\nu 2 \partial^\mu \partial^\nu \phi = 0$$

So the equation of motion becomes:

$$\partial_\mu \partial^\mu \partial_\nu \partial^\nu \phi = 0$$

From now on, for simplicity, I define:

$$\square := \partial_\mu \partial^\mu$$

So the equation of motion is of order 4:

$$\square^2 \phi = 0$$

An ordinary second-order differential equation can always be turned into two first-order equations by the following method:

$$f(x, \dot{x}, \ddot{x}) = 0$$

If I define:

$$v := \dot{x}$$

the equation takes the following form:

$$f(x, v, \dot{v}) = 0 \quad , \quad \dot{x} = v$$

In this way, two first-order equations were obtained that are completely equivalent to the second-order equation mentioned. Generalizing this method to rewriting the Lagrangian (rather than directly the equation of motion) is a little subtler; because when the Lagrangian is written in terms of two variables, these two variables are taken to be *independent* when extremizing the action. Therefore, in the example above, v must be taken to be independent, and $v = \dot{x}$ must be imposed through the equation of motion. I show this in our own problem. I want to write the action in terms of two independent fields ϕ, χ in such a way that, besides extremizing the same previous action (that is, $(\square\phi)^2$), it gives:

$$\chi = \square\phi$$

This can be done with Lagrange multipliers:

$$\tilde{\mathcal{L}}(\phi, \chi, \lambda) = \chi^2 + \lambda(\chi - \square\phi)$$

I write the equations of motion of this Lagrangian:

$$\psi = \square\phi \quad , \quad \lambda + 2\chi = 0 \quad , \quad \square\lambda = 0$$

Here, I substitute λ from the middle equation into the Lagrangian. Note that in general, one cannot substitute an equation of motion into the Lagrangian, but

for algebraic equations (which have no derivatives), no problem arises. Finally, the Lagrangian takes the following form:

$$\tilde{\mathcal{L}}(\phi, \chi) = \chi^2 - 2\chi(\chi - \square\phi) = -\chi^2 + 2\chi\square\phi \equiv -\chi^2 - 2\partial_\mu\chi\partial^\mu\phi$$

So I wrote the Lagrangian in terms of two independent fields whose equations of motion give:

$$\square\chi = 0 \quad , \quad \square\phi = \chi$$

which is the same previous fourth-order equation of motion.

1.7 Ghost in the Theory

I examine the Lagrangian of the previous part a little more. With the following change of variables:

$$\Phi := \frac{\phi + \chi}{\sqrt{2}} \quad , \quad \Psi := \frac{\phi - \chi}{\sqrt{2}}$$

which means:

$$\phi = \frac{\Phi + \Psi}{\sqrt{2}} \quad , \quad \chi = \frac{\Phi - \Psi}{\sqrt{2}}$$

I rewrite the terms of the Lagrangian:

$$-2\partial_\mu\chi\partial^\mu\phi = -\partial_\mu\Phi\partial^\mu\Phi + \partial_\mu\Psi\partial^\mu\Psi$$

and

$$-\chi^2 = -\frac{1}{2}(\Phi^2 + \Psi^2) + \Psi\Phi$$

so the Lagrangian becomes:

$$\mathcal{L} = -\partial_\mu\Phi\partial^\mu\Phi + \partial_\mu\Psi\partial^\mu\Psi - \frac{1}{2}(\Phi^2 + \Psi^2) + \Psi\Phi$$

It is clear that the signs of the kinetic terms of these fields differ from one another. By calculating the noncovariant Hamiltonian and the conservation of the Hamiltonian from the Poisson bracket, one can see that this leads to unstable equations of motion. I call such terms ghosts³ in the theory. Later I will require the theory to be ghost-free⁴.

1.8 Constructing the Theory

I require the theory to have one degree of freedom (which means that at most two derivatives appear in each term of the Lagrangian), to be Lorentz-invariant and ghost-free, and to have linear equations of motion. The objects with which I want to construct the Lagrangian are these:

$$\eta^{\mu\nu} \quad , \quad \partial_\mu \quad , \quad \phi$$

The only scalar Lagrangian made from these that is second order and quadratic in ϕ is as follows:

$$\mathcal{L} = \partial_\mu\phi\partial^\mu\phi - V(\phi)$$

where V is an arbitrary function. Note that $\phi\square\phi$ is equivalent to this Lagrangian (that is, it differs by a total derivative).

³ghost

⁴ghost-free

Chapter 2

Electromagnetism

In this chapter, I take the dynamical field of the problem to be a vector and show that Maxwell electromagnetism is the only linear Lorentz-invariant theory with gauge symmetry.

Questions:

1. Can these same things also be generalized to the Berry phase/connection/curvature in quantum mechanics?

2.1 Vector Theory

I take the dynamical variable of the problem to be the 4-vector A^μ . I write the objects in terms of which the Lagrangian can be expressed:

$$A^\mu, \quad \eta^{\mu\nu}, \quad \partial_\mu$$

I require the Lagrangian to be scalar and to have one degree of freedom (at most two derivatives), to be quadratic, to be ghost-free, and to have gauge symmetry. I show that the only Lagrangian that satisfies these is the Maxwell Lagrangian.

The only scalar term with zero derivatives is $A^\mu A_\mu$. There is no scalar term with one derivative (that is not a total derivative). The terms containing two derivatives are as follows:

$$(\partial_\mu A^\mu)(\partial_\nu A^\nu), \quad (\partial_\mu A^\nu)(\partial^\mu A_\nu), \quad (\partial_\mu A^\nu)(\partial_\nu A^\mu)$$

The remaining possible terms differ from these by a total derivative. On the other hand:

$$(\partial_\mu A^\mu)(\partial_\nu A^\nu) \equiv -A^\mu \partial_\mu \partial_\nu A^\nu \equiv (\partial_\nu A^\mu)(\partial_\mu A^\nu)$$

So among the three expressions written, the left and right ones are equivalent. Therefore, the most general Lagrangian we want is of the following form:

$$\mathcal{L} = \alpha(\partial_\mu A^\nu)(\partial^\mu A_\nu) + \beta(\partial_\mu A^\mu)(\partial_\nu A^\nu) + \gamma A_\mu A^\mu$$

Now I write the Helmholtz decomposition of A^μ :

$$A_\mu = \partial_\mu \phi + A_\mu^T \quad ; \quad \partial^\mu A_\mu^T = 0$$

Substituting this into the Lagrangian gives:

$$\begin{aligned} \mathcal{L} &= \alpha \left(\partial_\mu \partial^\nu \phi + \partial_\mu (A^T)^\nu \right) \left(\partial^\mu \partial_\nu \phi + \partial^\mu A_\nu^T \right) \\ &\quad + \beta \left(\partial^\mu \partial_\mu \phi + \partial^\mu A_\mu^T \right) \left(\partial^\nu \partial_\nu \phi + \partial^\nu A_\nu^T \right) \\ &\quad + \gamma \left(\partial_\mu \phi + A_\mu^T \right) \left(\partial^\mu \phi + (A^T)^\mu \right) \\ &\equiv \alpha \left((\Box \phi)^2 + (A^T)^\nu \Box A_\nu^T \right) + \beta (\Box \phi)^2 \\ &\quad + \gamma \left(-\phi \Box \phi + (A^T)^2 \right) \end{aligned}$$

As I showed in the previous chapter, the term $(\Box \phi)^2$ in the Lagrangian creates a ghost. Therefore, requiring the theory to be ghost-free gives:

$$\alpha + \beta = 0$$

So:

$$\mathcal{L} = \alpha \left((\partial_\mu A^\nu)(\partial^\mu A_\nu) - (\partial_\mu A^\mu)(\partial_\nu A^\nu) \right) + \gamma A_\mu A^\mu$$

Given the gauge symmetry of the theory:

$$A_\mu \rightarrow A_\mu + \partial_\mu \xi$$

one can construct a gauge-invariant quantity:

$$F_{\mu\nu} := \partial_\mu A_\nu - \partial_\nu A_\mu$$

which is an antisymmetric tensor. In terms of this tensor, the Lagrangian simplifies:

$$\begin{aligned} F_{\mu\nu} F^{\mu\nu} &= 2(\partial_\mu A_\nu)(\partial^\mu A^\nu) - 2(\partial_\mu A_\nu)(\partial^\nu A^\mu) \\ &\equiv 2(\partial_\mu A^\nu)(\partial^\mu A_\nu) - 2(\partial_\nu A^\nu)(\partial_\mu A^\mu) \end{aligned}$$

So the Lagrangian took the following form:

$$\mathcal{L} = \alpha F_{\mu\nu} F^{\mu\nu} + \gamma A_\mu A^\mu$$

which is called the Proca¹ Lagrangian. One can choose α arbitrarily:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} m^2 A_\mu A^\mu$$

This Lagrangian is a massive vector field. This Lagrangian appears naturally in superconductivity². Its equation of motion is as follows:

$$\partial_\mu F^{\mu\nu} + m^2 A^\nu = 0$$

¹Proca

²Superconductivity

which, for a stationary field:

$$\nabla^2 \mathbf{A} = m^2 \mathbf{A}$$

so:

$$\mathbf{A}(x) \sim e^{-mx}$$

so the magnetic field does not penetrate into the superconductor because the field becomes massive (the Meissner effect³). This is the Anderson–Higgs mechanism.

Note that I have not yet introduced the condition of gauge symmetry. I require that under the transformation:

$$A_\mu \rightarrow A'_\mu = A_\mu + \partial_\mu \Lambda$$

the equivalence class of the Lagrangian does not change. Under this transformation, one obtains:

$$F_{\mu\nu} \rightarrow F'_{\mu\nu} = F_{\mu\nu} + \partial_\mu \partial_\nu \Lambda - \partial_\nu \partial_\mu \Lambda = F_{\mu\nu}$$

and

$$A_\mu A^\mu \rightarrow A_\mu A^\mu + 2A^\mu \partial_\mu \Lambda$$

so the mass term is not invariant under the (local) gauge transformation. After this requirement, the Lagrangian takes the form

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

shown. So these requirements uniquely give the homogeneous Maxwell equations. To take the sources of the field into account, one can consider a 4-vector J^μ and couple the field to it:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + J^\mu A_\mu$$

I write the gauge transformation of this term:

$$J^\mu A_\mu \rightarrow J^\mu A_\mu + J^\mu \partial_\mu \Lambda \equiv J^\mu A_\mu - \Lambda \partial_\mu J^\mu$$

So requiring gauge symmetry gives:

$$\partial_\mu J^\mu = 0$$

which means conservation of charge.

2.2 Energy-Momentum Tensor

2.3 Covariant Derivative

³Meissner-Effect

Chapter 3

Linear Theory of Gravity

In this chapter, I take the dynamical field to be a symmetric tensor and derive the linear theory with gauge symmetry.

Questions:

1. We could have defined the gauge transformation in several ways. Including $\eta_{\mu\nu}\xi$ and $\partial_\mu\partial_\nu\xi$ and so on. What particular significance does the way I defined the gauge transformation here have?
2. In all of these, we write terms that are at most quadratic and take a linear combination of all of them. This is independent of the equations of motion becoming second order. What particular reason is behind this assumption? Or can this assumption be reduced to a weaker assumption?
3. Here I took the tensor $H_{\mu\nu}$ to be symmetric from the beginning. If we remove this assumption or examine an antisymmetric tensor, how much will these things change?

3.1 Tensor Field

I take the dynamical field to be the symmetric tensor $H_{\mu\nu}$. Again, I require the Lagrangian to be scalar, quadratic, and to have a single degree of freedom. The possible terms are as follows:

$$\partial_\epsilon H_{\mu\nu} \partial_\delta H_{\gamma\lambda} \eta^{\dots} \eta^{\dots} \eta^{\dots}$$

where each of the $\dots$'s must be filled with two indices. The 6 indices can be arranged in $6!$ ways. Since η is symmetric, swapping the superscript indices of one of them does not create a new case (the number of cases is divided by 2^3). Also, changing the order of multiplication of the η 's does not create a new case either (the number of cases is divided by $3!$). So the total number of cases that can be written is:

$$\frac{6!}{2^3 \times 3!} = 15$$

But the H 's are also symmetric, so swapping the indices $\mu\nu$ and $\gamma\lambda$ does not create a new case either. For simplicity, I denote μ or ν by A , $\gamma\lambda$ by B , ϵ by C ,

and δ by D . I also represent each term in the form

$$(\dots; \dots; \dots)$$

where two letters are placed in each $\dots$. I write the possible cases:

$$(AA; BB; CD)$$

$$(AA; BC; BD)$$

$$(AB; AB; CD)$$

$$(AB; AD; BC)$$

$$(AC; AB; BD)$$

$$(AC; AD; BB)$$

Only these 6 possible cases remained. On the other hand, all terms that turn into one another by renaming the letters are also duplicates. The renaming $A/B \rightarrow C/D$ is not allowed (because we have two A/B 's). So the renamings are only of the form $A \leftrightarrow B$ and $C \leftrightarrow D$. If both transformations are applied simultaneously, this means renaming the indices. Also, the transformation $C \leftrightarrow D$ is carried out by a total derivative. So terms that turn into one another under these two transformations are equivalent. It is clear that the second and sixth turn into one another under $A \leftrightarrow B$, and the fourth and fifth under $C \leftrightarrow D$. So the only independent cases that remain are these:

$$(AA; BB; CD)$$

$$(AA; BC; BD)$$

$$(AB; AB; CD)$$

$$(AB; AD; BC)$$

I write these terms:

$$T_1 := \partial_\epsilon H_\mu{}^\mu \partial^\epsilon H_\nu{}^\nu$$

$$T_2 := \partial_\epsilon H_\mu{}^\mu \partial^\nu H_\nu{}^\epsilon$$

$$T_3 := \partial_\epsilon H^{\mu\nu} \partial^\epsilon H_{\mu\nu}$$

$$T_4 := \partial_\nu H^{\mu\nu} \partial^\epsilon H_{\mu\epsilon}$$

I also define $H := H_\mu{}^\mu$. Now I require the theory to have the following gauge symmetry:

$$H_{\mu\nu} \rightarrow H'_{\mu\nu} = H_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$$

Under the gauge transformation, the four terms written have the following terms added to them, respectively:

$$\delta T_1 \equiv 4\partial_\epsilon \partial_\mu \xi^\mu \partial^\epsilon H \equiv -4\Box H \partial_\mu \xi^\mu$$

$$\begin{aligned} \delta T_2 &\equiv 2\partial_\epsilon \partial_\mu \xi^\mu \partial^\nu H_\nu{}^\epsilon + \partial_\epsilon H (\partial^\nu \partial_\nu \xi^\epsilon + \partial^\nu \partial^\epsilon \xi_\nu) \\ &\equiv -2\partial_\mu \xi^\mu \partial_\epsilon \partial_\nu H^{\epsilon\nu} - 2\partial_\nu \xi^\nu \Box H \end{aligned}$$

$$\begin{aligned}\delta T_3 &\equiv \partial^\epsilon H_{\mu\nu}(\partial_\epsilon \partial^\mu \xi^\nu + \partial_\epsilon \partial^\nu \xi^\mu) + \partial_\epsilon H^{\mu\nu}(\partial^\epsilon \partial_\mu \xi_\nu + \partial^\epsilon \partial_\nu \xi_\mu) \\ &\equiv 4\Box \xi_\nu \partial_\mu H^{\mu\nu}\end{aligned}$$

$$\begin{aligned}\delta T_4 &\equiv \partial^\epsilon H_{\mu\epsilon}(\partial_\nu \partial^\mu \xi^\nu + \partial_\nu \partial^\nu \xi^\mu) + \partial_\nu H^{\mu\nu}(\partial^\epsilon \partial_\mu \xi_\epsilon + \partial^\epsilon \partial_\epsilon \xi_\mu) \\ &\equiv -2\partial_\epsilon \xi^\epsilon \partial_\mu \partial_\nu H^{\mu\nu} + 2\partial_\nu H^{\mu\nu} \Box \xi_\mu\end{aligned}$$

So if I write the total Lagrangian as

$$\mathcal{L} = aT_1 + bT_2 + cT_3 + dT_4$$

and require gauge symmetry, I will have:

$$-4a - 2b = 0$$

$$-2b - 2d = 0$$

$$4c + 2d = 0$$

which gives:

$$b = -2a \quad , \quad c = -a \quad , \quad d = 2a$$

So I determined the Lagrangian up to a constant. Clearly, this constant is arbitrary. I set $a = 1/2$ (it does not matter, but this way it becomes identical to the Fierz–Pauli¹ Lagrangian). So the Lagrangian of the problem was given uniquely (up to a constant factor):

$$\mathcal{L} = \frac{1}{2}\partial_\mu H \partial^\mu H - \partial_\mu H \partial_\nu H^{\mu\nu} - \frac{1}{2}\partial_\epsilon H^{\mu\nu} \partial^\epsilon H_{\mu\nu} + \partial_\mu H^{\mu\nu} \partial^\epsilon H_{\nu\epsilon}$$

Now I want to write the equation of motion of this tensor field. Instead of Euler–Lagrange, I vary the Lagrangian. I write the variation of the Lagrangian term by term:

$$\begin{aligned}\delta \mathcal{L} &= \partial_\mu H \partial^\mu \delta H - \partial_\mu \delta H \partial_\nu H^{\mu\nu} - \partial_\mu H \partial_\nu \delta H^{\mu\nu} - \partial_\epsilon H^{\mu\nu} \partial^\epsilon \delta H_{\mu\nu} \\ &\quad + \partial_\mu H^{\mu\nu} \partial^\epsilon \delta H_{\nu\epsilon} + \partial^\epsilon H_{\nu\epsilon} \partial_\mu \delta H^{\mu\nu} \\ &\equiv -\delta H \Box H + \delta H \partial_\mu \partial_\nu H^{\mu\nu} + \delta H_{\mu\nu} \partial^\mu \partial^\nu H + \delta H_{\mu\nu} \Box H^{\mu\nu} \\ &\quad - \delta H_{\nu\epsilon} \partial^\epsilon \partial_\mu H^{\mu\nu} - \delta H_{\mu\nu} \partial^\mu \partial_\epsilon H^{\nu\epsilon} \\ &= \delta H_{\mu\nu} \left(-\eta^{\mu\nu} \Box H + \eta^{\mu\nu} \partial_\alpha \partial_\beta H^{\alpha\beta} + \partial^\mu \partial^\nu H + \Box H^{\mu\nu} \right. \\ &\quad \left. - \partial^\nu \partial_\alpha H^{\alpha\mu} - \partial^\mu \partial_\alpha H^{\nu\alpha} \right)\end{aligned}$$

So the equation of motion of this Lagrangian is obtained:

$$\Box H_{\mu\nu} - \partial_\mu \partial^\alpha H_{\alpha\nu} - \partial_\nu \partial^\alpha H_{\alpha\mu} + \partial_\mu \partial_\nu H + \eta_{\mu\nu} (\partial_\alpha \partial_\beta H^{\alpha\beta} - \Box H) = 0$$

I choose the gauge so that:

$$\partial^\mu H_{\mu\nu} - \frac{1}{2}\partial_\nu H = 0$$

¹Fierz-Pauli

I show that such a gauge certainly exists:

$$\partial^\mu (H_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu) - \frac{1}{2} \partial_\nu (H + 2\partial_\mu \xi^\mu) = 0$$

So:

$$\square \xi_\nu = -\partial^\mu H_{\mu\nu} + \frac{1}{2} \partial_\nu H$$

which is an inhomogeneous wave equation, and the boundary conditions of ξ do not matter either (the only thing I want from the gauge is to satisfy this differential equation). Since the wave equation has a Green's function, this equation certainly has a solution too.

Now, with this gauge, the equation of motion reduces to the following equation:

$$\square (H_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} H) = 0$$

So by defining

$$\bar{H}_{\mu\nu} := H_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} H$$

one obtains:

$$\square \bar{H}_{\mu\nu} = 0$$

Therefore, this theory predicts solutions in the form of tensor waves that propagate at the speed of light.

Now I also consider a tensor source for this field:

$$\mathcal{L}_{\text{source}} = H_{\mu\nu} J^{\mu\nu}$$

and again apply the requirement of gauge symmetry to the Lagrangian:

$$\begin{aligned} \delta \mathcal{L}_{\text{source}} &= J^{\mu\nu} \partial_\mu \xi_\nu + J^{\mu\nu} \partial_\nu \xi_\mu = (J^{\mu\nu} + J^{\nu\mu}) \partial_\mu \xi_\nu \\ &\equiv -\xi_\nu \partial_\mu (J^{\mu\nu} + J^{\nu\mu}) \end{aligned}$$

So if this source is isotropic:

$$\partial_\mu J^{\mu\nu} = 0$$

which is a conservation law. This tensor is determined as we choose (it depends on what things we want to be able to affect $H_{\mu\nu}$ in this theory). One idea is that for gravitational problems, in which mass (or equivalently, energy) is the source of the field, we take this to be the energy-momentum tensor. In this case, the Lagrangian takes the following form:

$$\mathcal{L} = \frac{1}{2} \partial_\mu H \partial^\mu H - \partial_\mu H \partial_\nu H^{\mu\nu} - \frac{1}{2} \partial_\epsilon H^{\mu\nu} \partial^\epsilon H_{\mu\nu} + \partial_\mu H^{\mu\nu} \partial^\epsilon H_{\nu\epsilon} + \frac{\kappa}{2} H_{\mu\nu} T^{\mu\nu}$$

where κ is the interaction coupling coefficient. Note that it was not necessary for the source of the field to be the energy-momentum tensor, nor was it necessary for this tensor to be symmetric (in that case, only its symmetric part was conserved), and it was also not necessary for it to be coupled linearly to the field in the Lagrangian; but this source would certainly end up in tensor form. For example, if I had taken the source to be a 4-vector such as J^μ and coupled it to the field in the Lagrangian through the term $H_{\mu\nu} \partial^\nu J^\mu$, the tensor $\partial^\nu J^\mu$ would still effectively have been taken as the source of the field.

Taking this term into account, the equation of motion (without fixing the gauge) takes the following form:

$$\square H_{\mu\nu} - \partial_\mu \partial^\alpha H_{\alpha\nu} - \partial_\nu \partial^\alpha H_{\alpha\mu} + \partial_\mu \partial_\nu H + \eta_{\mu\nu} (\partial_\alpha \partial_\beta H^{\alpha\beta} - \square H) = -\frac{\kappa}{2} T_{\mu\nu}$$

and in the “harmonic²” gauge, it becomes:

$$\square \bar{H}_{\mu\nu} = -\frac{\kappa}{2} T_{\mu\nu}$$

We introduced a tensor field and wrote the Lagrangian for its evolution and for matter. So the theory is complete here without needing any additional principle or assumption, and one can derive the equations of motion from it. The coefficient κ must be determined in the laboratory, but later I will calculate it from the Newtonian limit of gravity. Also note that the field $H^{\mu\nu}$ itself has an energy-momentum tensor too, which must also be taken into account on the right-hand side of the last equation. In this sense, this theory is nonlinear.

3.2 Equation of Motion of a Particle

Now I want to derive the equation of motion of a point mass from this theory. First, I must write the energy-momentum tensor of the point mass. I also neglect the nonlinear effect mentioned, for simplicity. In section 1.5, I calculated the energy-momentum tensor of a point mass:

$$T^{\mu\nu} = m \int d\tau \dot{x}^\mu \dot{x}^\nu \delta^4(x - x(\tau))$$

which, in a general parametrization, takes the following form:

$$T^{\mu\nu} = m \int d\lambda \frac{\dot{x}^\mu \dot{x}^\nu}{\sqrt{\eta_{\rho\sigma} \dot{x}^\rho \dot{x}^\sigma}} \delta^4(x - x(\lambda))$$

Now I write the Lagrangian corresponding to the interaction of the field and the particle:

$$\mathcal{L}_{\text{interaction}} = \frac{\kappa}{2} H_{\mu\nu} T^{\mu\nu} = m \int d\lambda \frac{\kappa}{2} \frac{H_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}{\sqrt{\eta_{\rho\sigma} \dot{x}^\rho \dot{x}^\sigma}}$$

So the Lagrangian corresponding to the particle is obtained:

$$\mathcal{L}_{\text{particle}} + \mathcal{L}_{\text{interaction}} = -m \int d\lambda \left(\sqrt{\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} - \frac{\kappa}{2} \frac{H_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}{\sqrt{\eta_{\rho\sigma} \dot{x}^\rho \dot{x}^\sigma}} \right)$$

So:

$$\mathcal{L}_{\text{particle}} + \mathcal{L}_{\text{interaction}} = -m \int d\lambda \frac{\eta_{\mu\nu} - (\kappa/2) H_{\mu\nu}}{\sqrt{\eta_{\rho\sigma} \dot{x}^\rho \dot{x}^\sigma}} \dot{x}^\mu \dot{x}^\nu$$

If I define:

$$g_{\mu\nu} := \eta_{\mu\nu} + \kappa H_{\mu\nu}$$

²Harmonic

to first order in H , I will have:

$$\sqrt{g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu} = \sqrt{\eta_{\mu\nu}\dot{x}^\mu\dot{x}^\nu} - \frac{\kappa}{2} \frac{H_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}{\sqrt{\eta_{\rho\sigma}\dot{x}^\rho\dot{x}^\sigma}}$$

which is the same expression I obtained for the Lagrangian. So to first order in H , I have:

$$\mathcal{L}_{\text{particle}} + \mathcal{L}_{\text{interaction}} = -m \int d\lambda \sqrt{-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu}$$

where $g_{\mu\nu} = \eta_{\mu\nu} + \kappa H_{\mu\nu}$. For convenience later, I define:

$$h_{\mu\nu} := \kappa H_{\mu\nu}$$

These calculations show that, to first order, the problem of gravity is a geodesic problem in the non-Lorentzian metric $g_{\mu\nu}$.

3.3 Gravity and Electromagnetism

Now I consider the interaction of gravity and the electromagnetic field:

$$\mathcal{L} = \frac{\kappa}{2} H_{\mu\nu} T^{\mu\nu} - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

I write the energy-momentum tensor of the electromagnetic field:

$$T^{\mu\nu} = F^\mu{}_\lambda F^{\nu\lambda} + \frac{1}{4} \eta^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}$$

So:

$$H_{\mu\nu} T^{\mu\nu} = H_{\mu\nu} F^\mu{}_\lambda F^{\nu\lambda} + \frac{1}{4} H F_{\alpha\beta} F^{\alpha\beta}$$

As a result:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{\kappa}{2} H_{\mu\nu} F^\mu{}_\lambda F^{\nu\lambda} + \frac{\kappa}{8} H F_{\alpha\beta} F^{\alpha\beta}$$

Now I want to see whether gravity here too, to first order in $H_{\mu\nu}$, is equivalent to changing the metric from $\eta_{\mu\nu}$ to $g_{\mu\nu}$ or not. So I consider the following expression:

$$\tilde{\mathcal{L}} := -\frac{1}{4} \sqrt{-g} F^{\mu\nu} F_{\alpha\beta} g_{\alpha\mu} g_{\beta\nu}$$

where $\sqrt{-g}$ (where g is the determinant of the metric) appears because of the transformation of d^4x in the action. I calculate it to first order:

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa H_{\mu\nu} = \eta_{\mu\alpha} (\delta^\alpha_\nu + \kappa \eta^{\alpha\beta} H_{\beta\nu})$$

So:

$$\begin{aligned} \sqrt{-g} &= \sqrt{-\eta} \left(1 + \frac{\kappa}{2} \eta^{\mu\nu} H_{\mu\nu} \right) + \mathcal{O}(H^2) \\ &= \sqrt{-g} = 1 + \frac{\kappa}{2} H + \mathcal{O}(H^2) \end{aligned}$$

Also, the product of the metrics is obtained:

$$\begin{aligned} g_{\alpha\mu}g_{\beta\nu} &= (\eta_{\alpha\mu} + \kappa H_{\alpha\mu})(\eta_{\beta\nu} + \kappa H_{\beta\nu}) \\ &= \eta_{\alpha\mu}\eta_{\beta\nu} + \kappa\eta_{\alpha\mu}H_{\beta\nu} + \kappa\eta_{\beta\nu}H_{\alpha\mu} + \mathcal{O}(H^2) \end{aligned}$$

So:

$$g_{\alpha\mu}g_{\beta\nu}F^{\mu\nu}F^{\alpha\beta} = F_{\mu\nu}F^{\mu\nu} + 2\kappa H_{\beta\nu}F_{\mu}{}^{\beta}F^{\mu\nu} + \mathcal{O}(H^2)$$

As a result:

$$\tilde{\mathcal{L}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{\kappa}{8}H F_{\mu\nu}F^{\mu\nu} + \frac{\kappa}{2}H_{\beta\nu}F_{\mu}{}^{\beta}F^{\mu\nu} + \mathcal{O}(H^2)$$

which is the same $\mathcal{L}$ obtained in the interaction of gravity and electromagnetism.

3.4 Calculation of κ

I wrote the Lagrangian corresponding to the motion of a particle in the previous section:

$$\mathcal{L}_{\text{particle}} + \mathcal{L}_{\text{interaction}} = -m \int d\lambda \sqrt{g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu}}$$

In section 4.5, I will show that extremizing this action leads to the following equation:

$$\ddot{x}^{\mu} + \Gamma_{\alpha\beta}^{\mu}\dot{x}^{\alpha}\dot{x}^{\beta} = 0$$

Also, in section 4.7, I will show that in the limit of low speeds (Newtonian), this equation gives:

$$-\kappa H_{00} = -2\Phi$$

which means:

$$H_{00} = \frac{2}{\kappa}\Phi(\mathbf{r})$$

where Φ is the gravitational potential. I write the equation of motion in the harmonic gauge:

$$\square \bar{H}_{\mu\nu} = -\frac{\kappa}{2}T_{\mu\nu}$$

Also, for a stationary mass, the only nonzero element of the energy-momentum tensor is $T_{00} = \rho$, where ρ is its mass density. Therefore, for a gravitostatic³ problem, I will have:

$$\nabla^2 \bar{H}_{00} = -\frac{\kappa}{2}\rho(\mathbf{r})$$

and the remaining elements of $\bar{H}_{\mu\nu}$ vanish from the equation of motion and the boundary conditions. On the other hand, from the definition

$$\bar{H}_{\mu\nu} = H_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}H$$

, I write:

$$\begin{aligned} \bar{H}_{0i} &= H_{0i} = 0 \\ \bar{H}_{ij} &= H_{ij} - \frac{1}{2}\delta_{ij}H = 0 \end{aligned}$$

³I'm not sure such a word is real!

So $H_{\mu\nu}$ is diagonal, and its spatial diagonal elements are equal too:

$$H_{ii} = \frac{1}{2}H$$

So:

$$\sum_{i=1}^3 H_{ii} = H + H_{00} = \frac{3}{2}H$$

where the positive sign is because of the metric and the definition of H . So:

$$H_{00} = \frac{1}{2}H$$

So:

$$H_{\mu\nu} = \text{diag}(H_{00}, H_{00}, H_{00}, H_{00})$$

So:

$$\bar{H}_{00} = H_{00} + \frac{1}{2}H = 2H_{00}$$

So the equation of motion of H_{00} is obtained:

$$2\nabla^2 H_{00} = -\frac{\kappa}{2}\rho(\mathbf{r})$$

which, by substituting H_{00} in terms of the gravitational potential, gives:

$$\nabla^2 \Phi = -\frac{\kappa^2}{8}\rho(\mathbf{r})$$

On the other hand, Newtonian mechanics says:

$$\nabla^2 \Phi = -4\pi G \rho(\mathbf{R})$$

So:

$$\kappa = \sqrt{32\pi G}$$

Now that the theory is complete, I write all the equations of motion.

- The equation of motion of $H_{\mu\nu}$ without fixing the gauge:

$$\square H_{\mu\nu} - \partial_\mu \partial^\alpha H_{\alpha\nu} - \partial_\nu \partial^\alpha H_{\alpha\mu} + \partial_\mu \partial_\nu H + \eta_{\mu\nu} (\partial_\alpha \partial_\beta H^{\alpha\beta} - \square H) = -\sqrt{8\pi G} T_{\mu\nu}$$

- The equation of motion of $h_{\mu\nu}$ without fixing the gauge:

$$\square h_{\mu\nu} - \partial_\mu \partial^\alpha h_{\alpha\nu} - \partial_\nu \partial^\alpha h_{\alpha\mu} + \partial_\mu \partial_\nu h + \eta_{\mu\nu} (\partial_\alpha \partial_\beta h^{\alpha\beta} - \square h) = -16\pi G T_{\mu\nu} \quad (3.1)$$

- The equation of motion of $\bar{H}_{\mu\nu}$ in the harmonic gauge:

$$\square \bar{H}_{\mu\nu} = -\sqrt{8\pi G} T_{\mu\nu}$$

- The equation of motion of $\bar{h}_{\mu\nu}$ in the harmonic gauge:

$$\square \bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu}$$

From now on, I call these “*Einstein’s equations*.”

Chapter 4

Geometry

Given the appearance of geometric definitions in the theory, in this section I first develop the geometric concepts independently of the theory and then try to follow the properties of these concepts in the theory.

One can look at the problem in two ways. The first view is that a manifold¹ has been placed in a larger flat space, and we examine the geometry of the points on its surface. And in the second view, there is no larger space, and everything there is in this very space. For example, for a two-dimensional creature placed on the surface of a sphere, the first view is from the perspective of a three-dimensional creature, and the second view is from the perspective of the ant itself. The entire difference between these two is that in the second view, I must assume that the covectors and the connection are given. But in the first view, one can define these from the position vector. Also, the second view **seems** more general. If the connection is accepted as a primitive quantity, it need not be symmetric in its lower indices. But when the connection is defined from the first view, it is necessarily symmetric.

Because of the structure of this chapter, I have marked the theorems and their proofs.

Questions

- Can every space be placed in a larger Euclidean space?

4.1 Differentiation and Connection

I denote the position vector of each point in space by $\mathbf{x}$, which is a one-to-one and smooth function of a number of quantities $\{x^i\}_i$ that I call “coordinates.” I define the basis vectors at the point $\mathbf{x}_0$:

$$\mathbf{e}_i(\mathbf{x}_0) := \left. \frac{\partial \mathbf{x}}{\partial x^i} \right|_{\mathbf{x}_0}$$

I define the tangent space:

$$\mathbb{T}(\mathbf{x}_0) := \text{span}\{\mathbf{e}_i(\mathbf{x}_0)\}_i$$

¹Manifold

I call a vector such as $\mathbf{v}$ a local vector around $\mathbf{x}_0$ if:

$$\mathbf{v} \in \mathbb{T}(\mathbf{x}_0)$$

The position vector of each point in the neighborhood of $\mathbf{x}_0$ is as follows:

$$\mathbf{x}(x_0^i + \delta x^i) = \mathbf{x}_0 + \left. \frac{\partial \mathbf{x}}{\partial x^i} \right|_{\mathbf{x}_0} \delta x^i = \mathbf{x}_0 + \delta x^i \mathbf{e}_i(\mathbf{x}_0)$$

So the relative position of each point near $\mathbf{x}_0$ with respect to it is a local vector around it. Consider a local vector field:

$$\mathbf{v}(\mathbf{x}) = v^i(\mathbf{x}) \mathbf{e}_i(\mathbf{x})$$

From now on, to simplify the notation, I do not write the position dependence unless it becomes necessary. I denote the duals of the basis vectors $\mathbf{e}_i$ (which I call covectors) by $\mathbf{e}^i$. So:

$$\mathbf{e}^i \mathbf{e}_j = \delta_j^i$$

So:

$$v^i = \mathbf{e}^i \mathbf{v}$$

Note that defining the dual space (and therefore the covectors) does not require defining an inner product. Now I write the derivative of this vector:

$$\partial_j \mathbf{v} = \partial_j v^i \mathbf{e}_i + v^i \partial_j \mathbf{e}_i$$

The second term on the right-hand side of the equality is not necessarily an element of $\mathbb{T}$. Therefore, if I define the “connection”:

$$\Gamma_{ji}^k := \mathbf{e}^k \partial_j \mathbf{e}_i$$

one can write:

$$\partial_j \mathbf{e}_i = \Gamma_{ij}^k \mathbf{e}_k + \Delta_{ij}^k \mathbf{f}_k$$

where $\mathbf{f}_k$ is a basis of the space perpendicular to $\mathbb{T}$.

One can object that the derivative of a vector means calculating it at two different (and neighboring) points and subtracting them. Whereas these two vectors do not belong to one tangent space so that they can be added. Therefore, the choice of connection actually determines our definition of adding vectors in each other’s neighborhood.

Theorem 1. If the space can be placed in a larger flat space, then the connection is certainly symmetric in its two lower indices.

Proof.

$$\partial_j \mathbf{e}_i = \frac{\partial^2 \mathbf{x}}{\partial x^i \partial x^j} = \frac{\partial^2 \mathbf{x}}{\partial x^j \partial x^i} = \partial_i \mathbf{e}_j$$

So:

$$\Gamma_{ji}^k = \Gamma_{ij}^k$$

□

Finally, I define the covariant derivative of the vector $\mathbf{v}$ as the projection of the derivative $\partial_i \mathbf{v}$ onto $\mathbb{T}$:

$$\begin{aligned}\nabla_i \mathbf{v} &:= \mathbf{e}_k \mathbf{e}^k \partial_i \mathbf{v} \\ \nabla_i v^k &:= (\nabla_i \mathbf{v})^k = \mathbf{e}^k \partial_i \mathbf{v} = \partial_i v^k + \Gamma_{ij}^k v^j\end{aligned}$$

Note that a nonzero connection is not a sign that the space is curved. In a two-dimensional flat space, the connection in polar coordinates is nonzero, although the space is intrinsically flat. One can also take the derivative of a covector:

$$\begin{aligned}\mathbf{v} &= v_i \mathbf{e}^i \in \mathbb{T}^* \\ \partial_i \mathbf{v} &= \partial_i v_j \mathbf{e}^j + v_j \partial_i \mathbf{e}^j\end{aligned}$$

On the other hand:

$$\Gamma_{ij}^k := \mathbf{e}^k \partial_i \mathbf{e}_j = \partial_i (\mathbf{e}^k \mathbf{e}_j) - \partial_i \mathbf{e}^k \mathbf{e}_j = -\partial_i \mathbf{e}^k \mathbf{e}_j$$

So I define its covariant derivative as follows:

$$\begin{aligned}\nabla_i \mathbf{v} &:= \partial_i \mathbf{v} \mathbf{e}_k \mathbf{e}^k \\ \nabla_i v_j &:= (\partial_i \mathbf{v})_j = \partial_i v_j \mathbf{e}_j = \partial_i v_j - \Gamma_{ij}^k v_k\end{aligned}$$

4.2 Metric and Levi-Civita Connection

Up to here, there was no need to define an inner product in the space. Now I assume that some kind of inner product has been defined in the space. For clarity, I write the properties of the inner product (these **seem** sufficient for my use):

- $\mathbb{T} \times \mathbb{T} \rightarrow \mathbb{F}$
- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
- $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$

I define the metric:

$$g_{ij} := \mathbf{e}_i \cdot \mathbf{e}_j$$

By the definition of the inner product, it is symmetric, and I assume it is invertible. I represent its inverse as follows:

$$g^{ij} := g_{ij}^{-1}$$

So:

$$g^{ik} g_{kj} = g^{ik} \mathbf{e}_k \cdot \mathbf{e}_j = \delta_j^i$$

One observes that the action of $g^{ik} \mathbf{e}_k \cdot$ and $\mathbf{e}^i$ on the basis vectors is equal, and both are $\mathbb{T} \rightarrow \mathbb{F}$ and linear. Therefore, they are equal:

$$\mathbf{e}^i = g^{ik} \mathbf{e}_k \cdot \quad (4.1)$$

So with the metric, one can obtain the covectors from the vectors.

Theorem 2. The connection of inner products that do not explicitly depend on position is unique and equal to the Levi-Civita connection, which is defined as follows:

$$\Gamma_{ik}^n := \frac{1}{2} g^{nj} (\partial_k g_{ij} + \partial_i g_{jk} - \partial_j g_{ki}) \quad (4.2)$$

Proof. I differentiate the metric:

$$\partial_k g_{ij} = \mathbf{e}_j \cdot (\partial_k \mathbf{e}_i) + \mathbf{e}_i \cdot (\partial_k \mathbf{e}_j)$$

It is right here that I used the fact that the inner product does not explicitly depend on position. For example, if I temporarily denote the inner product of $\mathbf{e}_i$ and $\mathbf{e}_j$ by $g(\mathbf{e}_i, \mathbf{e}_j, x)$, then if the definition of the inner product is explicitly a function of time, its derivative will be as follows:

$$\begin{aligned} \partial_k g(\mathbf{e}_i, \mathbf{e}_j, \mathbf{r}) &= \frac{\partial g}{\partial \mathbf{e}_i} \partial_k \mathbf{e}_j + \frac{\partial g}{\partial \mathbf{e}_j} \partial_k \mathbf{e}_i + \frac{\partial g}{\partial x^k} \\ &= \mathbf{e}_i \cdot \partial_k \mathbf{e}_j + \mathbf{e}_j \cdot \partial_k \mathbf{e}_i + \frac{\partial g}{\partial x^k} \end{aligned}$$

Now I multiply both sides by g^{nj} :

$$g^{nj} \partial_k g_{ij} = \Gamma_{ki}^n + g^{nj} \mathbf{e}_i \cdot \partial_k \mathbf{e}_j$$

I write this for two other permutations of ijk :

$$g^{nj} \partial_i g_{jk} = \Gamma_{ik}^n + g^{nj} \mathbf{e}_k \cdot \partial_i \mathbf{e}_j$$

$$g^{nj} \partial_j g_{ki} = g^{nj} \mathbf{e}_i \cdot \partial_j \mathbf{e}_k + g^{nj} \mathbf{e}_k \cdot \partial_j \mathbf{e}_i$$

The first+the second-the third, given $\Gamma_{ij}^k = \Gamma_{ji}^k$ and $\partial_i \mathbf{e}_j = \partial_j \mathbf{e}_i$, gives:

$$\Gamma_{ik}^n := \frac{1}{2} g^{nj} \left(\partial_k g_{ij} + \partial_i g_{jk} - \partial_j g_{ki} \right)$$

□

4.3 Curvature

Now I take two covariant derivatives of a local vector:

$$\begin{aligned} \nabla_i \nabla_j v^k &= (\partial_i \mathbf{e}_m \mathbf{e}^m \partial_j \mathbf{v})^k \\ &= \mathbf{e}^k \partial_i (\mathbf{e}_m \mathbf{e}^m \partial_j (v^n \mathbf{e}_n)) \\ &= \mathbf{e}^k \partial_i (\mathbf{e}_m \Gamma_{jn}^m v^n + \mathbf{e}_n \partial_j v^n) \\ &= \Gamma_{im}^k \Gamma_{jn}^m v^n + \partial_i \Gamma_{jn}^k v^n + \Gamma_{jn}^k \partial_i v^n + \Gamma_{jn}^k \partial_j v^n + \partial_i \partial_j v^k \end{aligned}$$

where the last three terms are symmetric in i, j . So:

$$[\nabla_i, \nabla_j] v^k = (\partial_i \Gamma_{jn}^k - \partial_j \Gamma_{in}^k + \Gamma_{im}^k \Gamma_{jn}^m - \Gamma_{jm}^k \Gamma_{in}^m) v^n$$

I define the Riemann curvature tensor²:

$$R_{nji}^k := \partial_i \Gamma_{jn}^k - \partial_j \Gamma_{in}^k + \Gamma_{im}^k \Gamma_{jn}^m - \Gamma_{jm}^k \Gamma_{in}^m \quad (4.3)$$

²Riemann-Curvature-Tensor

So:

$$[\nabla_i, \nabla_j]v^k = R^k_{nji}v^n$$

which means that the Riemann tensor can also be written in this way:

$$R^k_{nji} := \mathbf{e}^k[\nabla_i, \nabla_j]\mathbf{e}_n$$

Also, one can construct a rank-two tensor from it:

$$R_{nj} := R^j_{nji} = \partial_i \Gamma^j_{jn} - \partial_j \Gamma^j_{in} + \Gamma^j_{im} \Gamma^m_{jn} - \Gamma^j_{jm} \Gamma^m_{in}$$

which is the Ricci tensor³. Up to here, only the connection (rather than the metric) was needed. If a metric has been defined, one can also construct a scalar from these:

$$R := g^{nj}R_{nj}$$

which is the Ricci scalar⁴.

I define a flat space as a space in which $\partial_i \mathbf{e}_j \in \mathbb{T}$. That is, the basis vectors of all the points belong to one linear space.

Theorem 3. In a flat space, the Riemann curvature tensor is always zero.

Proof. I write the definition of the covariant derivative:

$$\nabla_i \mathbf{v} := \mathbf{e}_k \mathbf{e}^k \partial_i \mathbf{v} = v^j \mathbf{e}_k \mathbf{e}^k \partial_i \mathbf{e}_j + \mathbf{e}_k \mathbf{e}^k \mathbf{e}_j \partial_i v^j$$

Given that $\mathbf{e}_k \mathbf{e}^k \mathbf{e}_i = \mathbf{e}_i$ and that the flatness of the space implies

$$\partial_i \mathbf{e}_j = \Gamma^k_{ij} \mathbf{e}_k$$

one obtains:

$$\nabla_i \mathbf{v} = (\partial_i v^n + \Gamma^k_{in} v^j) \mathbf{e}_k = \mathbf{e}_j \partial_i v^j + v^j \partial_i \mathbf{e}_j = \partial_i \mathbf{v}$$

So the covariant derivative of a vector in flat space is the same as the partial derivative. So:

$$R^k_{nji} := \mathbf{e}^k[\nabla_i, \nabla_j]\mathbf{e}_n = \mathbf{e}^k[\partial_i, \partial_j]\mathbf{e}_n = 0$$

□

Now I show some mathematical properties of the curvature tensor.

Theorem 4. The Riemann curvature tensor is antisymmetric in its last two indices:

$$R^k_{nji} = -R^k_{nij}$$

Proof. From its definition, it is clear that:

$$R^k_{nji} = \mathbf{e}^k[\nabla_i, \nabla_j]\mathbf{e}_n = -\mathbf{e}^k[\nabla_j, \nabla_i]\mathbf{e}_n = -R^k_{nij}$$

□

³Ricci-Tensor

⁴Ricci-Scalar

Theorem 5. The first Bianchi⁵ identity:

$$R^k_{nji} + R^k_{inj} + R^k_{jin} = 0$$

Proof. I write the definition of the curvature tensor:

$$R^k_{nji} = \mathbf{e}^k [\nabla_i, \nabla_j] \mathbf{e}_n$$

On the other hand:

$$\nabla_i \mathbf{e}_j = \mathbf{e}_k \Gamma^k_{ij} = \nabla_j \mathbf{e}_i$$

So:

$$[\nabla_i, \nabla_j] \mathbf{e}_k + [\nabla_k, \nabla_i] \mathbf{e}_j + [\nabla_j, \nabla_k] \mathbf{e}_i = \nabla_i (\nabla_j \mathbf{e}_k - \nabla_k \mathbf{e}_j) + \dots = 0$$

So:

$$R^k_{nji} + R^k_{inj} + R^k_{jin} = 0$$

□

Theorem 6. The second Bianchi identity:

$$\nabla_l R^k_{nji} + \nabla_i R^k_{nlj} + \nabla_j R^k_{nil} = 0$$

Proof. I write the covariant derivative of the curvature tensor:

$$\nabla_l R^k_{nji} = \nabla_l \mathbf{e}^k [\nabla_i, \nabla_j] \mathbf{e}_n + \mathbf{e}^k \nabla_l [\nabla_i, \nabla_j] \mathbf{e}_n$$

The first derivative of the first term can be written in terms of the connection:

$$\nabla_l R^k_{nji} = -\Gamma^k_{lm} \mathbf{e}^m [\nabla_i, \nabla_j] \mathbf{e}_n + \mathbf{e}^k \nabla_l [\nabla_i, \nabla_j] \mathbf{e}_n$$

In terms of the curvature tensor:

$$\nabla_l R^k_{nji} = -\Gamma^k_{lm} R^m_{nji} + \mathbf{e}^k \nabla_l [\nabla_i, \nabla_j] \mathbf{e}_n$$

But the second term is not like this. However, given that:

$$\nabla_l [\nabla_i, \nabla_j] + \nabla_i [\nabla_j, \nabla_l] + \nabla_j [\nabla_l, \nabla_i] = [\nabla_l, \nabla_i] \nabla_j + [\nabla_j, \nabla_l] \nabla_i + [\nabla_i, \nabla_j] \nabla_l$$

one can write:

$$\begin{aligned} \nabla_l R^k_{nji} + \nabla_i R^k_{nlj} + \nabla_j R^k_{nil} &= -\Gamma^k_{lm} R^m_{nji} + R^k_{jim} \Gamma^m_{ln} \\ &\quad - \Gamma^k_{im} R^m_{nlj} + R^k_{ljm} \Gamma^m_{in} \\ &\quad - \Gamma^k_{jm} R^m_{nil} + R^k_{ilm} \Gamma^m_{jn} \end{aligned}$$

Clearly, the left-hand side of the equality is a tensor. Therefore, if it is zero in one coordinate system, it will be zero in all coordinate systems. Since **there is always a frame in which the connection is zero** (by theorem 15), this expression is zero. That is:

$$\nabla_l R^k_{nji} + \nabla_i R^k_{nlj} + \nabla_j R^k_{nil} = 0$$

□

⁵Bianchi

Theorem 7. The Riemann curvature tensor is antisymmetric in its first two indices:

$$R^{kn}{}_{ji} = -R^{nk}{}_{ji}$$

Proof. I define the curvature tensor with lower indices:

$$\begin{aligned} R_{knji} &:= g_{km} R^m{}_{nji} = \mathbf{e}_k \cdot [\nabla_i, \nabla_j] \mathbf{e}_n = \mathbf{e}_k \cdot \nabla_i \nabla_j \mathbf{e}_n - \mathbf{e}_k \cdot \nabla_j \nabla_i \mathbf{e}_n \\ &= \nabla_i (\mathbf{e}_k \cdot \nabla_j \mathbf{e}_n) - \nabla_j (\mathbf{e}_k \cdot \nabla_i \mathbf{e}_n) - \nabla_i \mathbf{e}_k \cdot \nabla_j \mathbf{e}_n + \nabla_j \mathbf{e}_k \cdot \nabla_i \mathbf{e}_n \end{aligned}$$

So:

$$R_{knji} + R_{nkji} = [\nabla_i, \nabla_j] (\mathbf{e}_k \cdot \mathbf{e}_n) = 0$$

It is also clear that if I raise the indices k and n , this identity still holds (because the covariant derivative of the metric is zero):

$$R^{kn}{}_{ji} = -R^{nk}{}_{ji}$$

□

Now I want to derive identities for the Ricci tensor. I obtain this by contracting⁶ the previous identities.

Theorem 8. The Ricci tensor is symmetric:

$$R_{ij} = R_{ji}$$

Proof. Contracting k and j in the first Bianchi identity gives:

$$R^j{}_{nji} + R^j{}_{inj} + R^j{}_{jin} = 0$$

The first term is zero because of the previous theorem. The second term also simplifies in terms of the Ricci tensor using this antisymmetry:

$$R_{ni} = R_{in}$$

which means that the Ricci tensor is symmetric. □

Theorem 9. The Einstein tensor, which is defined as follows:

$$G_{ij} := R_{ij} - \frac{1}{2} g_{ij} R$$

is symmetric, and its covariant derivative is zero:

$$\nabla_i G^{ij} = 0$$

Proof. Contracting k and j in the second Bianchi identity gives:

$$\nabla_l R^j{}_{nji} + \nabla_i R^j{}_{nlj} + \nabla_j R^j{}_{nil} = 0$$

where its first two terms can be written in terms of the Ricci tensor:

$$\nabla_l R_{ni} - \nabla_i R_{nl} + \nabla_j R^j{}_{nil} = 0$$

⁶contraction

Now, by contracting n and l

$$\nabla_n R^n{}_i - \nabla_i R + \nabla_j R^{jn}{}_{in} = 0$$

where the last term simplifies given $R^{jn}{}_{in} = R^{nj}{}_{ni} = R^j{}_i$:

$$\nabla_n R^n{}_i - \nabla_i R + \nabla_j R^j{}_i = 0$$

So:

$$\nabla_j (R^j{}_i - \frac{1}{2} \delta_i^j R) = 0$$

which, by raising the index i , becomes:

$$\nabla_j (R^{ji} - \frac{1}{2} g^{ij} R) = 0$$

So from the definition of the Einstein tensor:

$$\nabla_i G^{ij} = 0$$

Its symmetry is also clear from the symmetry of the metric and the Ricci tensor. $\square$

I list the properties that I showed in this section:

- Symmetries:

$$R^k{}_{nji} = -R^k{}_{nij} \quad , \quad R^{kn}{}_{ji} = -R^{nk}{}_{ji} \quad , \quad R_{ij} = R_{ji}$$

- The first Bianchi identity:

$$R^k{}_{nji} + R^k{}_{inj} + R^k{}_{jin} = 0$$

- The second Bianchi identity:

$$\nabla_l R^k{}_{nji} + \nabla_i R^k{}_{nlj} + \nabla_j R^k{}_{nil} = 0$$

- The Einstein tensor:

$$G_{ij} := R_{ij} - \frac{1}{2} g_{ij} R \quad , \quad \nabla_i G^{ij} = 0$$

4.4 Coordinate Transformation

4.4.1 Vectors

Now I want to examine the behavior of the quantities defined under a change of coordinates:

$$\{x^i\}_i \rightarrow \{x'^i\}_i$$

The coordinates of an n -dimensional space are a one-to-one map from the points of the space to $\mathbb{R}^n$. Since (by assumption) $\{x^i\}_i$ and $\{x'^i\}_i$ are both coordinates, a one-to-one function exists such that:

$$x'^i = f^i(x^1, x^2, \dots)$$

For simplicity of notation, I denote f by the same x' . I assume this function is continuous and differentiable. From symmetry, it is clear that it is invertible too. I write the definition of the basis vectors:

$$\mathbf{e}_i := \frac{\partial \mathbf{x}}{\partial x^i} \quad , \quad \mathbf{e}'_i := \frac{\partial \mathbf{x}}{\partial x'^i}$$

Since:

$$\frac{\partial}{\partial x'^i} = \frac{\partial x^j}{\partial x'^i} \frac{\partial}{\partial x^j}$$

so:

$$\mathbf{e}'_i = \frac{\partial x^j}{\partial x'^i} \mathbf{e}_j$$

and similarly:

$$\mathbf{e}_i = \frac{\partial x'^j}{\partial x^i} \mathbf{e}'_j$$

I call this type of transformation a covariant transformation. I also define:

$$\mathbf{e}'^i := \frac{\partial x'^i}{\partial x^j} \mathbf{e}^j$$

I call this type of transformation a contravariant transformation too. It is clear that:

$$\mathbf{e}'^i \mathbf{e}'_j = \frac{\partial x'^i}{\partial x^n} \frac{\partial x^n}{\partial x'^j} \delta_j^i = \delta_j^i$$

so the $\mathbf{e}'^i$'s are the covectors of the new basis vectors. Now one can see how the components of a local vector transform:

$$\mathbf{v} = v^i \mathbf{e}_i = v'^i \mathbf{e}'_i$$

To calculate v'^j , it is enough to apply $\mathbf{e}'^j$ to both sides:

$$v'^j = v^i \mathbf{e}'^j \mathbf{e}_i = v^i \frac{\partial x'^j}{\partial x^k} \mathbf{e}^k \mathbf{e}_i$$

So:

$$v'^j = \frac{\partial x'^j}{\partial x^i} v^i$$

So the components transform contravariantly. I call quantities with zero indices that do not change under coordinate transformations scalars. I call quantities with one index that transform covariantly/contravariantly covariant/contravariant vectors.

4.4.2 Tensors

I define a tensor by taking the tensor product of the basis vectors:

$$\mathbf{T} = T^{ij} \mathbf{e}_i \otimes \mathbf{e}_j$$

(I leave familiarity with its definition and properties to the reader). I find the transformation of the T^{ij} 's exactly as for a vector:

$$T'^{ij} = T^{nm} \mathbf{e}'^i \mathbf{e}_n \otimes \mathbf{e}'^j \mathbf{e}_m$$

So:

$$T'^{ij} = \frac{\partial x'^i}{\partial x^n} \frac{\partial x'^j}{\partial x^m} T^{nm}$$

It is clear that one can similarly define quantities of the form $T_{ij}, T^i_j, T_i^j, \dots$ too, and also with more indices. It is also clear from (4.1) that if a metric can be defined in the space, one can relate upper and lower indices to one another. For example:

$$T^i_j \mathbf{e}_i \otimes \mathbf{e}^j = T^i_j g^{jk} \mathbf{e}_i \otimes \mathbf{e}_k.$$

where the inner product in front of the second basis vector is because the second space is the dual space. But it still transforms like the basis vectors. Finally, it followed that:

$$T^{ik} = T^i_j g^{jk}$$

that is, it is as if “indices can be raised and lowered with the metric.” This statement is a little imprecise given the presence of the inner product mentioned, but it is consistent with the transformations.

Theorem 10. The metric is the components of a tensor.

Proof. I write the transformation of the metric:

$$g'_{ij} = \mathbf{e}'_i \cdot \mathbf{e}'_j = \frac{\partial x^n}{\partial x'^i} \frac{\partial x^m}{\partial x'^j} \mathbf{e}_n \cdot \mathbf{e}_m$$

So:

$$g'_{ij} = \frac{\partial x^n}{\partial x'^i} \frac{\partial x^m}{\partial x'^j} g_{nm}$$

So the following quantity is a tensor:

$$\mathbf{g} := g_{ij} \mathbf{e}^i \otimes \mathbf{e}^j = (\mathbf{e}_i \cdot \mathbf{e}_j) \mathbf{e}^i \otimes \mathbf{e}^j$$

□

Now I calculate the derivative of tensors:

$$\partial_k \mathbf{T} = \partial_k (T^{ij} \mathbf{e}_i \otimes \mathbf{e}_j) = \partial_k T^{ij} \mathbf{e}_i \otimes \mathbf{e}_j + T^{ij} \partial_k \mathbf{e}_i \otimes \mathbf{e}_j + T^{ij} \mathbf{e}_i \otimes \partial_k \mathbf{e}_j$$

I define the covariant derivative:

$$\nabla_k T^{ij} := (\partial_k \mathbf{T})^{ij} = (\mathbf{e}^i \otimes \mathbf{e}^j) \partial_k T$$

So:

$$\nabla_k T^{nm} = \partial_k T^{nm} + \Gamma_{ik}^n T^{im} + \Gamma_{jk}^m T^{nj}$$

It is also clear that:

$$\nabla_k T^n_m = \partial_k T^n_m + T^i_m \Gamma_{ik}^n - T^n_j \Gamma_{mk}^j$$

Theorem 11. If the inner product is not explicitly a function of position, the covariant derivative of its metric is zero:

$$\nabla_k g_{ij} = 0$$

Proof. I calculate the covariant derivative of the metric:

$$\begin{aligned}\nabla_k g_{ij} &= (\partial_k \mathbf{g})_{ij} = (\partial_k (\mathbf{e}_n \cdot \mathbf{e}_m \mathbf{e}^n \otimes \mathbf{e}^m))_{ij} \\ &= \partial_k \mathbf{e}_i \cdot \mathbf{e}_j + \mathbf{e}_i \cdot \partial_k \mathbf{e}_j + \mathbf{e}_n \cdot \mathbf{e}_j \partial_k \mathbf{e}^n \mathbf{e}_i + \mathbf{e}_i \cdot \mathbf{e}_n \partial_k \mathbf{e}^n \mathbf{e}_j \\ &= 0\end{aligned}$$

where in the last line I used $-\mathbf{e}^i \partial_k \mathbf{e}_j = \partial_k \mathbf{e}^i \mathbf{e}_j$. Therefore, the covariant derivative of the metric is zero. $\square$

Theorem 12. The covariant derivative of a tensor is a tensor.

Proof. I write the covariant derivative of the tensor $\mathbf{T}$:

$$\nabla'_k T'^{ij} := (\partial'_k \mathbf{T}')^{ij} = (\mathbf{e}'^i \otimes \mathbf{e}'^j)(\partial'_k \mathbf{T}')$$

Also, under a coordinate transformation:

$$\begin{aligned}\mathbf{e}'^i &= \frac{\partial x'^i}{\partial x^n} \mathbf{e}^n \\ \partial'_k &= \frac{\partial x^l}{\partial x'^k} \partial_l \\ \mathbf{T}' &= \mathbf{T}\end{aligned}$$

So:

$$\nabla'_k T'^{ij} = \frac{\partial x'^i}{\partial x^n} \frac{\partial x'^j}{\partial x^m} \frac{\partial x^l}{\partial x'^k} (\partial_l \mathbf{T})^{nm}$$

So:

$$\nabla'_k T'^{ij} = \frac{\partial x'^i}{\partial x^n} \frac{\partial x'^j}{\partial x^m} \frac{\partial x^l}{\partial x'^k} \nabla_l T^{nm}$$

so it is a tensor. (This proof holds for any number of indices) $\square$

Theorem 13. The partial derivative of a component of a tensor is not a component of a tensor.

Proof. I prove this for a vector (a rank-one tensor). For a rank- n tensor, the result is similar too, but instead of one extra term, it has n extra terms (arising from differentiation).

$$\partial'_i A'^j = \frac{\partial x^l}{\partial x'^k} \partial_l \left(\frac{\partial x'^i}{\partial x^n} A^n \right) = \frac{\partial x^l}{\partial x'^k} \frac{\partial x'^i}{\partial x^n} \partial_l A^n + \frac{\partial x^l}{\partial x'^k} \frac{\partial^2 x'^i}{\partial x^l \partial x^n} A^n$$

$\square$

Theorem 14. The connection is not a tensor and transforms as follows:

$$\Gamma'^k_{ij} = \frac{\partial x'^k}{\partial x^n} \frac{\partial x^m}{\partial x'^i} \frac{\partial x^l}{\partial x'^j} \Gamma^n_{ml} + \frac{\partial x'^k}{\partial x^n} \frac{\partial^2 x^n}{\partial x'^i \partial x'^j}$$

Proof.

$$\begin{aligned}\Gamma'^k_{ij} &= \mathbf{e}'^k \partial'_i \mathbf{e}'_j = \frac{\partial x'^k}{\partial x^n} \frac{\partial x^m}{\partial x'^i} \mathbf{e}^n \partial_m \left(\frac{\partial x^l}{\partial x'^j} \mathbf{e}_l \right) \\ &= \frac{\partial x'^k}{\partial x^n} \frac{\partial x^m}{\partial x'^i} \frac{\partial x^l}{\partial x'^j} \mathbf{e}^n \partial_m \mathbf{e}_l + \frac{\partial x'^k}{\partial x^n} \frac{\partial x^m}{\partial x'^i} \frac{\partial^2 x^n}{\partial x^m \partial x'^j}\end{aligned}$$

So:

$$\Gamma_{ij}'^k = \frac{\partial x'^k}{\partial x^n} \frac{\partial x^m}{\partial x'^i} \frac{\partial x^l}{\partial x'^j} \Gamma_{ml}^n + \frac{\partial x'^k}{\partial x^n} \frac{\partial^2 x^n}{\partial x'^i \partial x'^j}$$

□

Theorem 15. For any arbitrary point, one can find coordinates in which the connection is zero at that point.

Proof. content...

□

Now consider an arbitrary small change of coordinates:

$$x^\mu \rightarrow x'^\mu = x^\mu - \xi^\mu(x)$$

So:

$$x^\mu = x'^\mu + \xi^\mu$$

I write the transformation of the metric:

$$\frac{\partial x^\alpha}{\partial x'^\mu} = \delta_\mu^\alpha - \partial'_\mu \xi^\alpha$$

So:

$$g'_{\mu\nu}(x') = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} g_{\alpha\beta} = (\delta_\mu^\alpha + \partial'_\mu \xi^\alpha)(\delta_\nu^\beta + \partial'_\nu \xi^\beta) g_{\alpha\beta}(x)$$

which, to first order, gives:

$$g'_{\mu\nu} = g_{\mu\nu} + g_{\mu\alpha} \partial'_\nu \xi^\alpha + g_{\alpha\nu} \partial'_\mu \xi^\alpha$$

Substituting the metric, I write:

$$g'_{\mu\nu} = g_{\mu\nu} + \partial'_\nu \xi_\mu + \partial'_\mu \xi_\nu$$

The similarity of this transformation to the gauge transformation examined in the previous chapter is clear.

4.4.3 Tensor Densities

4.5 Geodesic

Now I want to find the curve of shortest length for an arbitrary metric. From the definition of the metric, the length is calculated:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

So:

$$s = \int \sqrt{g_{\mu\nu} dx^\mu dx^\nu} = \int \sqrt{g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda$$

where λ is an arbitrary parameter for traversing the path in question. I want this integral to be extremized. One can write Euler–Lagrange with the following Lagrangian:

$$L = \sqrt{g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} \quad (4.4)$$

$$\frac{\partial L}{\partial x^\alpha} - \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^\alpha} \right) = 0$$

where $\dot{x} := dx/d\lambda$. But instead of doing this, for simplicity, I vary the Lagrangian:

$$\begin{aligned} \delta L &= \frac{1}{2L} \left(\delta g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu + 2g_{\mu\nu} \dot{x}^\mu \delta \dot{x}^\nu \right) \\ &\equiv \frac{1}{2L} \delta x^\alpha \partial_\alpha g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu - \delta x^\nu \frac{d}{d\lambda} \left(\frac{1}{L} \dot{x}^\mu g_{\mu\nu} \right) \\ &= \delta x^\nu \left(\frac{1}{2L} \partial_\nu g_{\mu\alpha} \dot{x}^\mu \dot{x}^\alpha - \frac{1}{L} \ddot{x}^\mu g_{\mu\nu} - \frac{1}{L} \partial_\alpha g_{\mu\nu} \dot{x}^\alpha \dot{x}^\mu + \frac{1}{L^2} \frac{dL}{d\lambda} g_{\mu\nu} \dot{x}^\mu \right) \end{aligned}$$

So I obtained the equation of the shortest path (geodesic⁷):

$$g_{\mu\nu} \ddot{x}^\mu - \frac{1}{2} \partial_\nu g_{\mu\alpha} \dot{x}^\mu \dot{x}^\alpha + \partial_\alpha g_{\mu\nu} \dot{x}^\alpha \dot{x}^\mu = \frac{1}{L} \frac{dL}{d\lambda} g_{\mu\nu} \dot{x}^\mu$$

Multiplying both sides by $g^{\beta\nu}$, I will have:

$$\ddot{x}^\beta + g^{\beta\nu} \left(\partial_\alpha g_{\mu\nu} - \frac{1}{2} \partial_\nu g_{\mu\alpha} \right) \dot{x}^\mu \dot{x}^\alpha = \frac{1}{L} \frac{dL}{d\lambda} \dot{x}^\beta$$

Since the second term on the left-hand side of the equality is symmetric in the indices μ and α , I can also write it in the following more symmetric form:

$$\ddot{x}^\beta + g^{\beta\nu} \left(\frac{\partial_\alpha g_{\mu\nu} + \partial_\mu g_{\alpha\nu}}{2} - \frac{1}{2} \partial_\nu g_{\mu\alpha} \right) \dot{x}^\mu \dot{x}^\alpha = \frac{1}{L} \frac{dL}{d\lambda} \dot{x}^\beta$$

From the definition of the Levi-Civita connection (4.2), the equation takes the following form:

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = \frac{dx^\mu}{d\lambda} \frac{d}{d\lambda} \ln L$$

Now I want to change the parametrization of the path from λ to s , where:

$$ds = L d\lambda$$

This parametrization is called an affine parameter⁸. So:

$$\begin{aligned} L \frac{d}{ds} &= \frac{d}{d\lambda} \\ \frac{d^2 x^\mu}{d\lambda^2} &= L \frac{d}{ds} \left(L \frac{dx^\mu}{ds} \right) = L^2 \frac{d^2 x^\mu}{ds^2} + L \frac{dL}{ds} \frac{dx^\mu}{ds} \end{aligned}$$

In this way, I can rewrite the geodesic equation:

$$L^2 \frac{d^2 x^\mu}{ds^2} + L \frac{dL}{ds} \frac{dx^\mu}{ds} + L^2 \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{ds} \frac{dx^\beta}{ds} = L \frac{dx^\mu}{ds} \frac{dL}{ds}$$

which simplifies to:

$$\frac{d^2 x^\mu}{ds^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{ds} \frac{dx^\beta}{ds} = 0 \quad (4.5)$$

⁷Geodesic

⁸Affine-Parameter

This equation is the geodesic equation. This parametrization has another property too. Given:

$$\frac{ds}{d\lambda} = L(\lambda)$$

and

$$L(s) = \sqrt{g_{\mu\nu} \frac{dx^\mu}{ds} \frac{dx^\nu}{ds}} = \frac{d\lambda}{ds} \sqrt{g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} = \frac{d\lambda}{ds} L = 1$$

in fact, for any parameter of the form

$$\lambda = a s + b$$

which I call the family of affine parameters, its equation of motion will be (4.5). It is clear that this result works both ways. Also, since ds is an imaginary massive particle, let me write it in terms of proper time ($ds = i d\tau$):

$$L(\tau)^2 = -1$$

Now I want to see how the equation of motion changes if I remove the square root from the Lagrangian (4.4):

$$\tilde{L}(\lambda) = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \quad (4.6)$$

I extremize the action of this Lagrangian:

$$\begin{aligned} \delta \tilde{s} &= \int \left(\delta g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu + 2g_{\alpha\nu} \delta \dot{x}^\alpha \dot{x}^\nu \right) d\lambda \\ &\equiv \int \left(\frac{\partial g_{\mu\nu}}{\partial x^\alpha} \dot{x}^\mu \dot{x}^\nu - 2g_{\alpha\nu} \ddot{x}^\nu - 2 \frac{\partial g_{\alpha\nu}}{\partial x^\mu} \dot{x}^\mu \dot{x}^\nu \right) \delta x^\alpha d\lambda = 0 \end{aligned}$$

So:

$$\ddot{x}^\alpha + g^{\alpha\beta} \left(\frac{\partial g_{\beta\nu}}{\partial x^\mu} - \frac{1}{2} \frac{\partial g_{\mu\nu}}{\partial x^\beta} \right) \dot{x}^\mu \dot{x}^\nu = 0$$

which, as before, by symmetrizing the first term and defining the Levi-Civita connection, gives:

$$\ddot{x}^\alpha + \Gamma_{\mu\nu}^\alpha \dot{x}^\mu \dot{x}^\nu = 0$$

which is the same equation (4.5), but there is no condition on the parametrization. Therefore, the solutions of the equation of motion of the Lagrangian (4.6), with the condition

$$L^2 = -1$$

imposed, are the same solutions of the equation of motion arising from (4.4). This point is useful because the equation of motion of (4.6) is usually simpler.

4.6 Velocity and Acceleration

Now I want to define velocity and acceleration for the motion of a particle in this space:

$$\mathbf{v} := \frac{d\mathbf{x}}{dt} = \dot{x}^i \partial_i \mathbf{x} = \dot{x}^i \mathbf{e}_i$$

So if I define

$$v^i := \dot{x}^i$$

then:

$$\mathbf{v} = v^i \mathbf{e}_i$$

To define acceleration, I differentiate this expression:

$$\mathbf{a} := \frac{d\mathbf{v}}{dt} = \dot{v}^i \mathbf{e}_i + v^i v^j \partial_j \mathbf{e}_i$$

So:

$$\mathbf{e}^k \mathbf{a} = a^k = \dot{v}^k + \Gamma_{ij}^k v^i v^j$$

which means:

$$a^k = \frac{d^2 x^k}{dt^2} + \Gamma_{ij}^k \frac{dx^i}{dt} \frac{dx^j}{dt}$$

Therefore, the geodesic equation is the same as zero acceleration in the comoving frame.

4.7 Geometric Version of Newtonian Gravity

The geodesic equation and zero acceleration are both of the following form:

$$\ddot{x}^\mu + \Gamma_{\alpha\beta}^\mu \dot{x}^\alpha \dot{x}^\beta = 0$$

If the speed of the particle is much less than the speed of light, the largest term of this equation is as follows:

$$\ddot{x}^\mu + \Gamma_{00}^\mu = 0$$

Since:

$$\Gamma_{00}^\mu = -\frac{1}{2} g^{\mu\nu} \partial_\nu g_{00}$$

if the metric is slightly perturbed relative to Minkowski:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

then:

$$\Gamma_{00}^\mu = -\frac{1}{2} \eta^{\mu\nu} \partial_\nu h_{00} = -\frac{1}{2} \partial^\mu h_{00}$$

so:

$$\ddot{x}^\mu = \frac{1}{2} \partial^\mu h_{00}$$

whose spatial components give:

$$\ddot{\mathbf{x}} = \frac{1}{2} \nabla h_{00}$$

so if I choose:

$$h_{00} = -2\Phi(\mathbf{x})$$

where $\Phi(\mathbf{x})$ is the gravitational potential, the equation of motion is obtained:

$$\ddot{\mathbf{x}} = -\nabla\Phi$$

which is the same as Newtonian gravity. Note that in this method, the equality of inertial mass and gravitational mass was not brought up at all. That is, this holds naturally here.

Chapter 5

Developing Linear Gravity

In this chapter, I develop the theory and examine various problems.

5.1 SVT Decomposition

The following combination appears in both the geometric and algebraic approaches:

$$ds^2 := g_{\mu\nu} dx^\mu dx^\nu = (\eta_{\mu\nu} + h_{\mu\nu}) dx^\mu dx^\nu$$

I separate its spatial and time parts:

$$h_{00} =: -2\Phi$$

$$h_{0i} =: A_i$$

So:

$$ds^2 = -(1 + 2\Phi) dt^2 + 2A_i dx^i dt + (\delta_{ij} + h_{ij}) dx^i dx^j$$

Any vector such as A_i can be Helmholtz-decomposed:

$$A_i = \partial_i A + A_i^T, \quad \partial^i A_i^T = 0$$

Now I want to decompose a tensor such as h_{ij} . I define:

$$v_j := \partial^i h_{ij}$$

I write the Helmholtz decomposition of this vector:

$$\partial^i h_{ij} = \partial_j f + v_j^T, \quad \partial^j v_j^T = 0$$

Now I define B_j as the solution of the following Laplace equation:

$$\nabla^2 B_j := v_j^T$$

Also, since:

$$\nabla^2 \partial^j B_j = \partial^j v_j^T = 0$$

so:

$$\partial^j B_j = 0$$

now I define from it:

$$h_{ij}^{(1)} := h_{ij} - \partial_i B_j - \partial_j B_i$$

I differentiate:

$$\partial^i h_{ij}^{(1)} = \partial_j f + v_j^T - \nabla^2 B_j = \partial_j f$$

So if I define B as the solution of the following Laplace equation:

$$\nabla^2 B := f$$

and define:

$$h_{ij}^T := h_{ij}^{(1)} - \partial_i \partial_j B$$

one obtains:

$$\partial^i h_{ij}^T = \partial_j f - \partial_j \nabla^2 B = 0$$

and similarly:

$$\partial^j h_{ij}^T = 0$$

therefore, I decomposed the tensor h_{ij} as follows:

$$h_{ij} = \partial_i B_j + \partial_j B_i + \partial_i \partial_j B + h_{ij}^T, \quad \partial^i h_{ij}^T = 0$$

The trace part of any tensor such as T_{ij} can also be separated as follows:

$$T_{ij} = \frac{1}{3} T_k^k \delta_{ij} + (T_{ij} - \frac{1}{3} T_k^k \delta_{ij})$$

where the first term is a scalar and the second term is a traceless¹ tensor. I do this for h_{ij}^T . I define:

$$h_{ij}^{TT} := h_{ij}^T - \frac{1}{3} h_k^k \delta_{ij} - \partial_i B_j - \partial_j B_i - (\partial_i \partial_j - \frac{1}{3} \delta_{ij} \nabla^2) B$$

This is a traceless tensor. Finally, by defining $-2\Psi := 1/3 h_k^k$, the decomposition of h_{ij} was obtained as follows:

$$h_{ij} = -2\Psi \delta_{ij} + \partial_i B_j + \partial_j B_i + (\partial_i \partial_j - \frac{1}{3} \delta_{ij} \nabla^2) B + h_{ij}^{TT}$$

where:

$$\partial^i B_i = 0$$

$$\partial^i h_{ij}^{TT} = \partial^j h_{ij}^{TT} = 0$$

$$h^{TT}{}_{,i}{}^i = 0$$

The decomposition of the metric $h_{\mu\nu}$ in terms of the scalars A, B, Φ, Ψ , the vectors A_i^T and B_i^T , and the tensor h_{ij}^{TT} is called scalar-vector-tensor decomposition²(SVT).

¹Traceless

²Scalar-Vector-Tensor

5.2 Transformations of the Decomposition Coefficients

Note that each of these is a scalar, vector, or tensor with respect to the metric $\eta_{\mu\nu}$. Therefore, their transformations under coordinate/gauge transformations with respect to the metric $g_{\mu\nu}$ are not necessarily scalar, vector, or tensor transformations. In what follows, I write their transformations. The gauge transformation is given by ξ_μ . For simplicity, I define:

$$T := \xi^0, \quad L_i := \xi_i$$

where one can also perform a Helmholtz decomposition on L^i :

$$L_i = \partial_i L + L_i^T$$

In section 4.4.2, I obtained the transformations of $h_{\mu\nu}$ under these transformations:

$$h'_{\mu\nu} = h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$$

Therefore:

$$-2\Phi' = h'_{00} = -2\Phi + 2\partial_0 \xi_0 = -2\Phi - 2\dot{T}$$

So:

$$\Phi' = \Phi + \dot{T}$$

Also:

$$A'_i = h'_{0i} = h_{0i} + \partial_0 \xi_i + \partial_i \xi_0 = h_{0i} + \dot{L}_i - \partial_i T$$

So:

$$\partial_i A' + A_i^{T'} = \partial_i A + A_i^T + \partial_i \dot{L} + \dot{L}_i^T - \partial_i T$$

I differentiate both sides:

$$\nabla^2 (A' - A - \dot{L} + T) = 0$$

With boundary conditions at infinity, one obtains:

$$A' = A + \dot{L} - T$$

Substituting this again, I will have:

$$A_i^{T'} = A_i^T + \dot{L}_i^T$$

Now I go to the transformations of the parts corresponding to h_{ij} :

$$h'_{ij} = h_{ij} + \partial_i L_j + \partial_j L_i$$

I take the trace of both sides:

$$h'^i_i = -\frac{2}{3}\Psi' = -\frac{2}{3}\Psi + 2\partial^i L_i = -\frac{2}{3}\Psi + 2\nabla^2 L$$

So:

$$\Psi' = \Psi - \frac{1}{3}\nabla^2 L$$

Now, instead of taking the trace, I take ∂^i of both sides:

$$\partial^i h'_{ij} = -2\partial_j \Psi' + \nabla^2 B'_j + \frac{2}{3}\partial_j \nabla^2 B' = \partial^i h_{ij} + \nabla^2 L_j^T + 2\partial_j \nabla^2 L$$

Substituting Ψ' into this gives:

$$\frac{2}{3}\partial_j \nabla^2 L + \nabla^2 B'_j + \frac{2}{3}\partial_j \nabla^2 B' = \nabla^2 B_j + \frac{2}{3}\partial_j \nabla^2 B + \nabla^2 L_j^T + 2\partial_j \nabla^2 L$$

I take ∂^j of both sides:

$$\frac{2}{3}\nabla^4(B' - B - 2L) = 0$$

So:

$$B' = B + 2L$$

Substituting this into the equation before taking ∂^j gives:

$$\nabla^2(B'_j - B_j - L_j^T) = 0$$

So:

$$B'_j = B_j + L_j^T$$

And finally, by rewriting the initial transformation equation and substituting all of these, one obtains:

$$h_{ij}^{TT'} = h_{ij}^{TT'}$$

Now that I have obtained all the transformations, I write them all together (also, since $\partial^i B_i = 0$, I denote it by B_i^T):

- Scalars:

$$\Phi' = \Phi + \dot{T}$$

$$\Psi' = \Psi - \frac{1}{3}\nabla^2 L$$

$$A' = A + \dot{L} - T$$

$$B' = B + 2L$$

- Vectors:

$$A_i^{T'} = A_i^T + \dot{L}_i^T$$

$$B_i^{T'} = B_i^T + L_i^T$$

- Tensors:

$$h'_{ij} = h_{ij}$$

As is clear, the transformations of each type have been completely separated from one another. Given these transformations, one can construct four gauge-invariant quantities (or, in the geometric approach, scalars under small coordinate changes with respect to the metric $g_{\mu\nu}$).

$$\Phi + \dot{A} - \ddot{B}$$

$$\Psi + \frac{1}{6}\nabla^2 B$$

$$A_i^T - \dot{B}_i^T$$

$$h_{ij}^{TT}$$

where there are two scalars with one degree of freedom, one vector with two degrees of freedom, and one tensor with two degrees of freedom.

5.3 Gravity of a Point Mass

The energy-momentum tensor for a massive fluid is as follows:

$$T^{00} = \rho \quad , \quad T^{0i} = (\rho + p)v^i \quad , \quad T^{ij} = (\rho + p)v^i v^j + \delta^{ij}p$$

where, if the source is static and pressureless, its only nonzero element will be $T^{00} = \rho$. For this energy-momentum tensor, I write Einstein's equations:

$$\square \bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu}$$

So for $\mu = \nu = 0$, it gives:

$$\square \bar{h}_{00} = -16\pi G \rho$$

and the right-hand side of the equality is zero for the remaining matrix elements. Since I want the static case, I can replace the $\square$'s with ∇^2 :

$$\nabla^2 \bar{h}_{00} = -16\pi G \rho$$

and the remaining matrix elements satisfy Laplace's equation and are zero at infinity. The uniqueness theorem for solutions of Laplace's equation (with boundary conditions) requires these to be zero everywhere. So:

$$\begin{aligned} \bar{h}_{00} &= 4G \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}' =: -4\Phi(\mathbf{r}) \\ \bar{h} &= 4\Phi \end{aligned}$$

On the other hand:

$$\bar{h}_{\mu\nu} := h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h \quad , \quad h = -\bar{h}$$

So:

$$h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\bar{h}$$

which gives:

$$h_{00} = -2\Phi$$

$$h_{i0} = 0$$

$$h_{ij} = -2\Phi\delta_{ij}$$

This is the same result I also reached in section 3.4:

$$h_{00} = -2\Phi(\mathbf{r}) \quad , \quad h_{\mu\nu} = h_{00}\text{diag}(1, 1, 1, 1)$$

For a point mass (or at points far from the mass), this takes the following form (I take $G = 1$):

$$\Phi(\mathbf{R}) = -\frac{m}{R}$$

which is the same Newtonian potential. Therefore:

$$ds^2 = -\left(1 - \frac{2m}{R}\right)dt^2 + \left(1 + \frac{2m}{R}\right)d\mathbf{R}^2$$

which, in spherical spatial coordinates:

$$ds^2 = - \left(1 - \frac{2m}{R}\right) dt^2 + \left(1 + \frac{2m}{R}\right) (dR^2 + R^2 d\Omega^2)$$

if our problem is perturbative (in m/R), one can completely separate the angular part with a change of variables:

$$r^2 := R^2 \left(1 + \frac{2m}{R}\right)$$

Note that the zeroth order of r is the same as R . So:

$$r^2 = R^2 \left(1 + \frac{2m}{R}\right) + \mathcal{O}\left(\left(\frac{m}{R}\right)^2\right)$$

$$R = r \left(1 - \frac{m}{r}\right) + \mathcal{O}\left(\left(\frac{m}{r}\right)^2\right)$$

So to first order, one obtains:

$$dR = dr$$

So the metric takes the following form to first order:

$$ds^2 = - \left(1 - \frac{2m}{r}\right) dt^2 + \left(1 + \frac{2m}{r}\right) dr^2 + r^2 d\Omega^2$$

5.4 Path of Light

The path of light is a geodesic path in which $ds^2 = 0$. The geodesic paths can be calculated from Euler–Lagrange:

$$S = \int ds = \int L(\lambda) d\lambda$$

$$L^2 = - \left(1 - \frac{2m}{r}\right) \dot{t}^2 + \left(1 + \frac{2m}{r}\right) \dot{r}^2 + r^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2)$$

where the dot means a derivative with respect to the parameter λ along the path. It is clear that because the sign of ds^2 does not change along the path and x^2 is a one-to-one and increasing function, one can extremize the integral of L^2 instead of L . This only makes the algebra simpler. So:

$$\frac{\partial L^2}{\partial x} - \frac{d}{d\lambda} \left(\frac{\partial L^2}{\partial \dot{x}} \right) = 0 \quad , \quad x \in \{t, r, \theta, \phi\}$$

I write it for t, θ, ϕ . For t :

$$\frac{d}{d\lambda} \left[\left(1 - \frac{2m}{r}\right) \dot{t} \right] = 0$$

$$\dot{t} = \frac{B}{1 - 2m/r}$$

where c is a constant. For θ :

$$r^2 \sin \theta \cos \theta \dot{\phi}^2 - \frac{d}{d\lambda} (r^2 \dot{\theta}) = 0$$

For ϕ :

$$\frac{d}{d\lambda} (r^2 \sin^2 \theta \dot{\phi}) = 0$$

I choose the coordinates so that at the initial moment, $\theta = \pi/2$ and $\dot{\theta} = 0$. The θ equation gives:

$$\frac{d}{d\lambda} (r^2 \dot{\theta}) = 0$$

So at all times, $\theta = \pi/2$. The ϕ equation gives:

$$\frac{d}{d\lambda} (r^2 \dot{\phi}) = 0$$

which is conservation of angular momentum:

$$r^2 \dot{\phi} = A$$

Also, light follows the path $L^2 = 0$. For now, I do not substitute it. So:

$$L^2 + \frac{B^2}{1 - 2m/r} = \left(1 + \frac{2m}{r}\right) \dot{r}^2 + r^2 \dot{\phi}^2$$

$$B^2 + L^2 \left(1 - \frac{2m}{r}\right) = \dot{r}^2 + \left(1 - \frac{2m}{r}\right) \frac{A^2}{r^2}$$

where in the first line I substituted $\dot{t}$, and in the second line $\dot{\phi}$. I want to calculate $r(\theta)$. I apply the following change of variables:

$$u := \frac{1}{r}$$

$$\dot{r} = -\frac{\dot{u}}{u^2} = -\frac{1}{u^2} \frac{du}{d\phi} \frac{A}{r^2} = -A \frac{du}{d\phi}$$

So the differential equation of the path is as follows:

$$A^2 \left(\frac{du}{d\theta}\right)^2 + (1 - 2mu)(A^2 u^2 - L^2) - B^2 = 0$$

For simplicity, I differentiate once:

$$A^2 \frac{d^2 u}{d\theta^2} + (1 - 2mu)A^2 u - m(A^2 u^2 - L^2) = 0$$

So:

$$\frac{d^2 u}{d\theta^2} + u = 3mu^2 - \frac{mL^2}{A^2}$$

For light, $L = 0$:

$$\frac{d^2 u}{d\theta^2} + u = 3mu^2$$

I solve this perturbatively:

$$u = u^{(0)} + mu^{(1)}$$

Zeroth order gives:

$$\frac{d^2 u^{(0)}}{d\theta^2} + u^{(0)} = 0$$

So:

$$u^{(0)} = \frac{1}{b} \cos(\theta - \theta_0)$$

where b is a constant. I choose the origin of θ so that $\theta_0 = 0$:

$$r = \frac{b}{\cos \theta}$$

which is the equation of a vertical straight line whose minimum distance from the origin is b . Now I write the first order:

$$\frac{d^2 u^{(1)}}{d\theta^2} + u^{(1)} = 3(u^{(0)})^2 = \frac{3}{b^2} \cos^2 \theta$$

which becomes:

$$\frac{d^2 u^{(1)}}{d\theta^2} + u^{(1)} = \frac{3}{2b^2} (\cos(2\theta) + 1)$$

The solution of this equation is as follows:

$$u^{(1)} = \frac{1}{b^2} \left(\frac{3 - \cos(2\theta)}{2} \right)$$

So:

$$u(\theta) = \frac{1}{b} \cos \theta + \frac{m}{b^2} \left(\frac{3 - \cos(2\theta)}{2} \right)$$

I take the angle at which $u = 0$ (which means $r \rightarrow \infty$) to be ϕ . Clearly, its zeroth order is $\pi/2$. I calculate its first order from this equation:

$$\phi^{(1)} = \frac{2}{b} = 0$$

So:

$$\phi = \frac{\pi}{2} + \frac{2m}{b}$$

Because of the $\pm\phi$ symmetry of the solutions of the equation, the total deflection angle is twice this amount. Also, I measure the “deflection” from the angle π . So:

$$\Delta\theta = \frac{4m}{b}$$

which, in terms of the speed of light and Newton’s constant (which I took to be one), becomes:

$$\Delta\theta = \frac{4Gm}{c^2 b}$$

For the Sun, one obtains:

$$\frac{4Gm}{c^2} \approx 5.9\text{km}$$

If I take b (which in this problem was the minimum distance of light from the origin) to be the radius of the Sun, one obtains:

$$\frac{4Gm}{c^2 R} \approx 1.751''$$

which predicts well the same value of $1.75''$ that Eddington observed in 1919 [1].

5.5 Path of a Particle

Consider a planet moving in the same metric that was calculated in 5.3. Its path is also geodesic, but its L^2 is always negative. In 4.5, I showed that one can always find a parameter along the path such that $L^2 = -1$. So all the equations are as before, except that instead of $L^2 = 0$, I must substitute $L^2 = -1$. I put this into the equations of the previous section:

$$\frac{d^2 u}{d\theta^2} + u = 3mu^2 + \frac{m}{A^2}$$

which, in terms of c and G , becomes:

$$\frac{d^2 u}{d\theta^2} + u = \frac{Gm}{A^2} + \frac{3Gm}{c^2} u^2$$

where the coefficient of the second term on the right is small:

$$\epsilon := \frac{3Gm}{c^2}$$

I solve the problem perturbatively by taking it to be small. Zeroth order is the Newtonian elliptical orbit:

$$\begin{aligned} \frac{d^2 u^{(0)}}{d\theta^2} + u^{(0)} &= \frac{Gm}{A^2} \\ u^{(0)}(\theta) &= a \cos \theta + \frac{Gm}{A^2} \end{aligned}$$

Direct Perturbation

Its first order can be written from the equation:

$$\frac{d^2 u^{(1)}}{d\theta^2} + u^{(1)} = u_0^2 = a^2 \cos^2 \theta + \frac{G^2 m^2}{A^4} + \frac{2Gma}{A^2} \cos \theta$$

So:

$$\frac{d^2 u^{(1)}}{d\theta^2} + u^{(1)} = u_0^2 = \frac{a^2}{2} + \frac{a^2}{2} \cos(2\theta) + \frac{G^2 m^2}{A^4} + \frac{2Gma}{A^2} \cos \theta$$

Its particular solution becomes:

$$u^{(1)} = \frac{a^2}{2} + \frac{G^2 m^2}{A^4} - \frac{a^2}{6} \cos(2\theta) + \frac{Gma}{A^2} \cos \theta$$

So to first order:

$$u(\theta) = a \cos \theta - \frac{3Gma^2}{6c^2} \cos(2\theta) + \frac{3G^2m^2a}{c^2A^2} \theta \cos \theta + \frac{Gm}{A^2} + \frac{3Gma^2}{2c^2} + \frac{3G^3m^3}{c^2A^4}$$

The last three terms are constant. The first two terms have a period of 2π , but clearly the third term disrupts this period. Therefore, its orbit should not be closed. But this solution is only valid for small θ 's because it diverges and the perturbation assumption breaks down.

Lindstedt–Poincare Method

Instead of the previous method, to keep the perturbation from diverging, I apply the following change of variables:

$$\phi := \frac{\theta}{k}$$

where k is an arbitrary function of ϵ . I take it to first order as follows:

$$k(\epsilon) := 1 + \epsilon k^{(1)} + \mathcal{O}(\epsilon^2)$$

and I will determine $k^{(1)}$ so that the divergence of the solution is eliminated. Given

$$\frac{d^2}{d\theta^2} = \frac{1}{k^2} \frac{d^2}{d\phi^2}$$

the equation of the path takes the following form:

$$\frac{1}{k^2} u'' + u = \frac{Gm}{A^2} + \epsilon u^2$$

where the derivatives are with respect to ϕ . To zeroth order in ϵ , we have $k = 1$, so:

$$u^{(0)''} + u = \frac{Gm}{A^2}$$

whose solution is as follows:

$$u^{(0)}(\phi) = a \cos \phi + \frac{Gm}{A^2}$$

Now I write the equation to first order:

$$u^{(1)''} - 2k^{(1)}u^{(0)''} + u^{(1)} = (u^{(0)})^2$$

Substituting $u^{(0)}$ gives:

$$u^{(1)''} + 2k^{(1)}a \cos \phi + u^{(1)} = \frac{a^2}{2} + \frac{a^2}{2} \cos(2\phi) + \left(\frac{Gm}{A^2}\right)^2 + 2\frac{Gma}{A^2} \cos \phi$$

Clearly, the term that caused the solution to diverge is the last term on the right-hand side of the equality. I use my freedom in choosing k :

$$k^{(1)} := \frac{Gm}{A^2}$$

In this way, the divergent term was eliminated. I write the final equation:

$$u^{(1)''} + u^{(1)} = \frac{a^2}{2} + \frac{a^2}{2} \cos(2\phi) + \left(\frac{Gm}{A^2}\right)^2$$

whose particular solution is as follows:

$$u^{(1)} = \frac{a^2}{8} \cos(2\phi) + \frac{a^2}{2} + \left(\frac{Gm}{A^2}\right)^2$$

So to first order, one obtained:

$$u(\phi) = a \cos(\phi) + \epsilon \frac{a^2}{8} \cos(2\phi) + \frac{Gm}{A^2} + \epsilon \left(\frac{Gm}{A^2}\right)^2 + \epsilon \frac{a^2}{2} + \mathcal{O}(\epsilon^2)$$

If I take all the additional constants to be C , one obtains:

$$u(\theta) = a \cos\left((1 - k^{(1)}\epsilon)\theta\right) + \epsilon \frac{a^2}{8} \cos(2\theta) + C + \mathcal{O}(\epsilon^2)$$

although the phrase “to first order” is not very meaningful here, because the first term itself includes all odd orders of ϵ . But this was necessary for the convergence of the answer. Clearly, the second and third terms of this expression have a period of 2π , but the first term has a period of $2\pi(1 + \epsilon)$. Therefore, the perihelion of the planet’s orbit must rotate by

$$\delta\theta \approx 2\pi k^{(1)}\epsilon = \frac{6\pi G^2 m^2}{A^2 c^2}$$

in each revolution. One can also write it in terms of the orbital “eccentricity” if one wishes. From the definitions of mechanics:

$$\frac{Gm}{A^2} = \frac{1}{a(1 - e^2)}$$

So:

$$\delta\theta \approx \frac{6\pi Gm}{ac^2(1 - e^2)}$$

For the Sun:

$$\frac{6\pi Gm}{c^2} \approx 28\text{km} \approx 1.9 \times 10^{-7} \text{AU}$$

Therefore, this effect is very weak. Clearly, for planets closer to the Sun, this effect will be stronger. For Mercury, Venus, and Earth:

$$\delta\theta_{\text{Mercury}} \approx 5.0 \times 10^{-7} \text{rad/orbit}$$

$$\delta\theta_{\text{Venus}} \approx 2.6 \times 10^{-7} \text{rad/orbit}$$

$$\delta\theta_{\text{Earth}} \approx 1.9 \times 10^{-7} \text{rad/orbit}$$

all of which are of order 10^{-7} radians per revolution. But clearly, for circular orbits (that is, small e ’s), this effect (even if it exists) is not detectable because of its azimuthal symmetry. Given that:

$$e_{\text{Mercury}} \approx 0.206 \quad , \quad e_{\text{Venus}} \approx 0.007 \quad , \quad e_{\text{Earth}} \approx 0.017$$

therefore, this effect is observable only in Mercury. This effect was observed in Mercury’s orbit in 1859 AD (around 1237 in the Solar Hijri calendar), and its disagreement with the perturbations arising from the gravitational effects of Venus and Jupiter was confirmed. This observation in [2] confirms the value obtained in the calculations above.

References

- [1]: F. W. Dyson, A. S. Eddington, and C. Davidson, "Determination of the Deflection of Light by the Sun's Gravitational Field", *Philos. Trans. R. Soc. A* 220, 291–333 (1920).
- [2]: Le Verrier, U. (1859): "Lettre de M. Le Verrier à M. Faye sur la théorie de Mercure et sur le mouvement du périhélie de cette planète", *Comptes rendus hebdomadaires des séances de l'Académie des sciences*, 49, 379–383 (Paris, 1859).

Chapter 6

Einstein's Equations

6.1 Geometric Dynamical Equations

Continuing the correspondence between geometry and the theory of gravity, I try to calculate the curvature of the perturbed metric to first order:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

The Christoffel symbols to first order are these:

$$\Gamma_{\mu\nu}^{\alpha} = \frac{1}{2}\eta^{\alpha\beta}(\partial_{\mu}h_{\nu\beta} + \partial_{\nu}h_{\mu\beta} - \partial_{\beta}h_{\mu\nu})$$

So the last two terms of the Riemann tensor (4.3) vanish to first order. So to first order:

$$R^{\alpha}{}_{\mu\beta\nu} = \partial_{\beta}\Gamma_{\mu\nu}^{\alpha} - \partial_{\nu}\Gamma_{\mu\beta}^{\alpha}$$

In terms of the metric:

$$R^{\alpha}{}_{\mu\beta\nu} = \frac{1}{2}(-\partial_{\mu}\partial_{\beta}h_{\nu}^{\alpha} + \partial_{\mu}\partial_{\nu}h_{\beta}^{\alpha} - \partial^{\alpha}\partial_{\nu}h_{\beta\mu} + \partial^{\alpha}\partial_{\beta}h_{\nu\mu})$$

The Ricci tensor:

$$R_{\mu\nu} := R^{\alpha}{}_{\mu\alpha\nu} = \frac{1}{2}(-\partial_{\mu}\partial^{\alpha}h_{\alpha\nu} + \partial_{\mu}\partial_{\nu}h - \partial_{\nu}\partial^{\alpha}h_{\alpha\mu} + \square h_{\mu\nu})$$

The Ricci scalar:

$$R := \eta^{\mu\nu}R_{\mu\nu} = -\partial_{\mu}\partial_{\nu}h^{\mu\nu} + \square h$$

Comparison with (3.1) immediately gives:

$$R_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}R = -8\pi G T_{\mu\nu}$$

These are correct to first order. If I want to write the equation of motion only in terms of the geometry of the problem (the metric), it will be:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -8\pi G T_{\mu\nu} \quad (6.1)$$

This is the same Einstein tensor that I defined in theorem 9:

$$G_{\mu\nu} = -8\pi G T_{\mu\nu}$$

I emphasize again that these equations agree with the theory of gravity written in chapter three only when they are written to first order. The only advantage of this formulation at present is that I have turned the problem into a completely geometric problem.

Another advantage of this formulation is that energy conservation is more apparent in it. From theorem 9, one immediately proves:

$$\nabla_\mu T^{\mu\nu} = 0$$

One can also add another term to the equation of motion in such a way that it still satisfies energy conservation:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = -8\pi G T_{\mu\nu}$$

because the covariant derivative of the metric is zero by theorem 11. I call Λ the cosmological constant. Given that its linear limit is not the same theory obtained in the linear gravity chapter, the cosmological constant (if it exists) must be very small to preserve the successes of the previous theory.

One can also write Einstein's equation only in terms of the Ricci tensor. By contracting

$$R_{\mu\nu} + \left(-\frac{1}{2}R + \Lambda\right) g_{\mu\nu} = -8\pi G T_{\mu\nu}$$

one obtains:

$$R = 4\Lambda - 8\pi G T_\mu{}^\mu$$

Substituting into the previous equation:

$$R_{\mu\nu} = \Lambda g_{\mu\nu} - 8\pi G \left(T_{\mu\nu} + \frac{1}{2}T g_{\mu\nu}\right)$$

which, in vacuum, gives:

$$R_{\mu\nu} = \Lambda g_{\mu\nu}$$

These equations alone are not enough to calculate the metric. Einstein's equations, whether in vacuum or in the presence of energy, whether with the cosmological constant or without it, are a second-order differential equation. Therefore, boundary conditions are needed to solve it. Like electromagnetism, I can usually take the boundary conditions so that at points far from the presence of energy, the metric becomes Minkowski.

6.2 Homogeneous and Isotropic Solution

The homogeneous and isotropic metric becomes:

$$ds^2 = A(t) dt^2 - B(t) dr^2$$

That is:

$$g_{\mu\nu} = (A(t), -B(t), -B(t), -B(t))$$

$$g^{\mu\nu} = (A^{-1}(t), -B^{-1}(t), -B^{-1}(t), -B^{-1}(t))$$

One can calculate the Levi-Civita connection for it:

$$\begin{aligned}\Gamma_{00}^0 &= \frac{\dot{A}}{2A} \quad , \quad \Gamma_{i0}^0 = 0 \\ \Gamma_{00}^n &= 0 \quad , \quad \Gamma_{i0}^n = \frac{\dot{B}}{B} \delta_i^n \\ \Gamma_{ij}^n &= 0 \quad , \quad \Gamma_{ij}^0 = \frac{\dot{B}}{2A} \delta_{ij}\end{aligned}$$

With this connection, I calculate the Ricci tensor:

$$\begin{aligned}R_{00} &= -\frac{3\ddot{B}}{2B} + \frac{3\dot{B}^2}{4B^2} + \frac{3\dot{A}\dot{B}}{4AB} \\ R_{ij} &= \left(\frac{\ddot{B}}{2A} + \frac{\dot{B}^2}{4AB} \right) \delta_{ij}\end{aligned}$$

The Ricci scalar also becomes:

$$R = -\frac{3\ddot{B}}{AB} + \frac{3\dot{A}\dot{B}}{2A^2B}$$

6.3 Isotropic Solution

In this section, I first take the cosmological constant to be zero. Then I show its role. For zero cosmological constant, Einstein's equations take the following form:

$$R_{\mu\nu} = 0$$

Clearly, the Minkowski metric satisfies it. Now I want to calculate the gravitational field of a symmetric point mass again. From symmetry, I take the metric to be as follows:

$$ds^2 = (-1 + \Phi(r)) dt^2 + (1 + h(r)) dr^2$$